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LESSON 141

Examples of Reproduction in Reptiles and Birds

Reptiles are true terrestrial vertebrates. They have shelled eggs with extraembryonic membranes. Chorion and amnion give protection and provide aqueous environment. Allantois plays its role in gaseous exchange and to store excretory products. Reptiles contain large amount of yolk. Complete development takes place within shelled egg and there is no larval stage. Some snakes are ovoviviparous because they keep eggs within body and young ones are hatched out.

All birds except water fowl lack penis. They deposit semen against cloaca and fertilization is internal. Fertilization before shell formation. Hard shelled eggs are incubated by parents. There is more parental care and feeding than amphibians, reptiles and mammals.

LESSON 142

Examples of Reproduction in Mammals

Among mammals monotremes (duckbilled platypus, spiny anteater) lay eggs. Duckbilled platypus lay eggs in nest/burrows while spiny anteater incubate eggs in ventral pouch. Marsupial mammals like koala and kangaroo keep the young one in ventral pouch (marsupium). They have short gestation period (8- 40 days) while feeding from mammary glands (suckling) is for 60 to 120 days. Most of the mammals are placental mammals (deer, lion, monkeys etc.). They have internal development and long gestation period. After birth young one is nourished through mammary glands. The longest gestation period is seen in elephants which lasts for 22 months.

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LESSON 143

Reproductive Patterns in Animals

All invertebrates and vertebrates have one of the three reproductive patterns.

a. Oviparous

These are the animals which lay eggs and development takes place outside the body of animals. Fertilization may be internal or external. Majority of invertebrates and many vertebrates are oviparous.

b. Ovoviparous

These are the animals which retain eggs into the body but the young one gets all its food from yolk of egg. Both fertilization and development is internal. Many members of annelids, brachiopods, gastropods, fishes and reptiles.

c. Viviparous

These are the animals in which eggs develop into the body and the young one gets its nourishment from mother. Both fertilization and development is internal. Scorpions, most mammals, elasmobranch fishes, few amphibians and reptiles are viviparous.

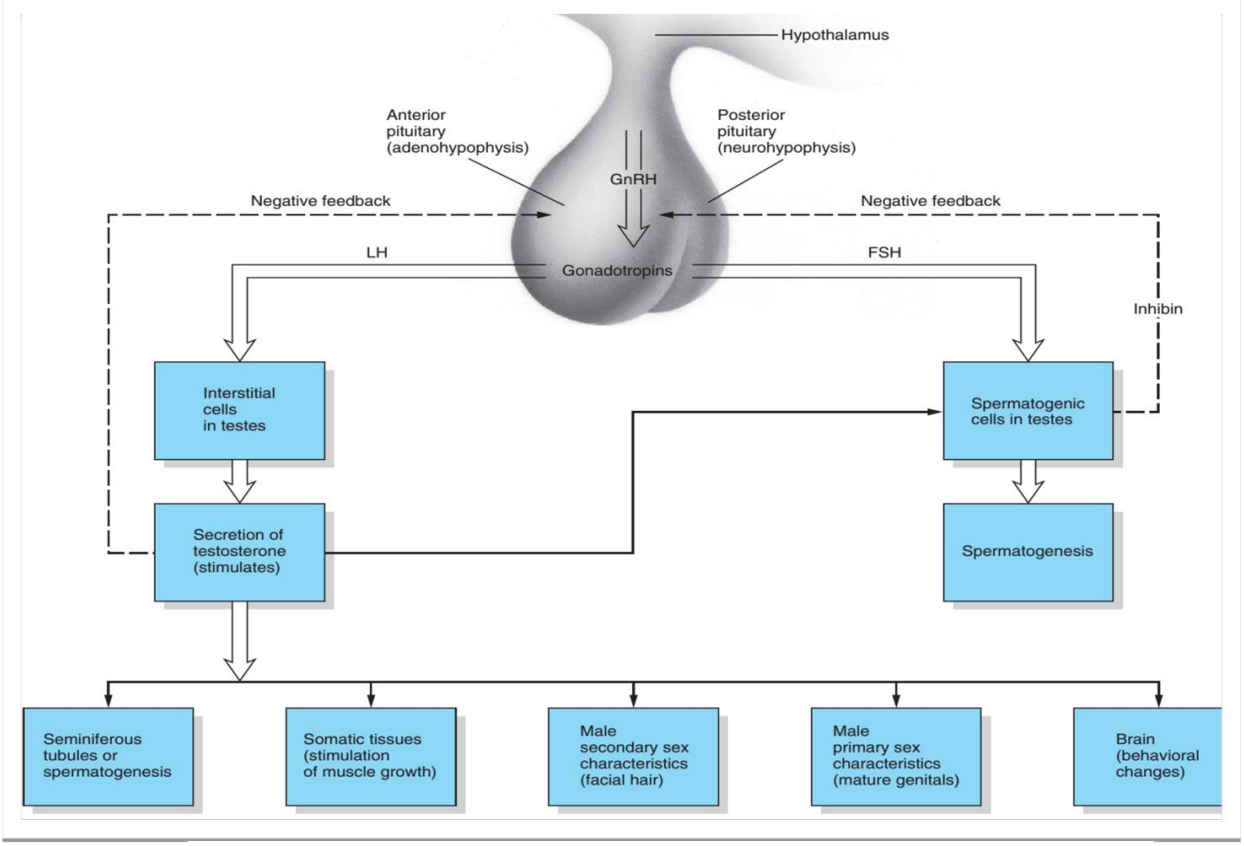


Fig. Hormonal control of male reproductive function

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LESSON 144

INVERTEBRATE REPRODUCTIVE SYSTEM

Invertebrates may have a simple reproductive structure which release gametes into water for external fertilization. Other may have complex structures for internal fertilization. Some invertebrates like polychaetes have no permanent reproductive organs. Cells lining body cavity produce gametes which are released through coelomic or nephridial duct. Sometimes body wall may rupture. However most of the vertebrates have male and female reproductive organs.

Male reproductive system is meant for production and safe transfer of sperms. Male reproductive system has testes for sperm production. Sperm pass through sperm duct/vas deferens to seminal vesicle for storage. Then sperms are passed out through ejaculatory duct. Accessory glands add seminal fluid.

Female reproductive system has ovaries where egg or ova is produced. Mature eggs pass through oviduct to a common genital chamber, from where they enter into female bursa (vagina). Then eggs are stored in seminal receptacles. In some species, sperms from one mating are stored and used all the life.

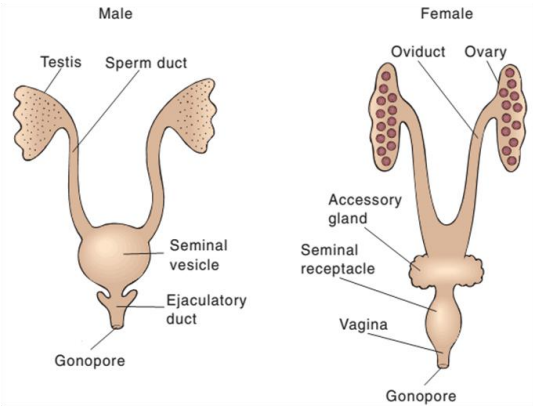


Fig. generalized structure of invertebrate reproductive system

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LESSON 145

Human Male Reproductive System

Male reproductive system have two basic function: Production of sperms and Transfer of sperm to female reproductive tract. Important components of male reproductive system include scrotum, testes, rete testis, epididymis, duct deferens and ampulla. Seminal Vesicle. Male reproductive system consists of a pair of testes enclosed in a sac like structure called scrotum. Each testes has a large number of seminiferous tubules. These are the sites for sperm production. Seminiferous tubule join a network of tubules called rete testis. After that there is a dilated tubule called epididymis. Epididymis has three functions: 1) storage and maturation of sperms. 2) Smooth muscles that help to propel sperm during ejaculation and 3) Transfer sperms to duct deferens. Duct deferens transfer comes out of scrotum and passes through the lower part of abdominal wall via inguinal canal. Sometime this point is weakened and intestine part may protrude out. This condition is called inguinal hernia. Around urinary bladder duct deferens enlarges to form ampulla. Next to ampulla is called ejaculatory duct. After combining with urinary tube it moves in penis as urethra.

There are different glands present in reproductive system. Important glands are seminal vesicles, prostate gland and bulbourethral gland.

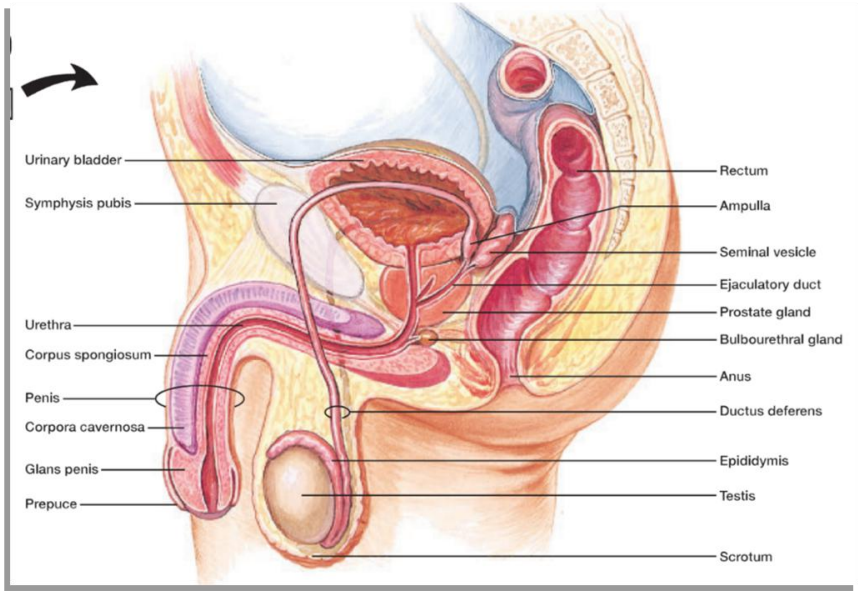


Fig. The human male reproductive system

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LESSON 146

Production of Sperms in human

Sperm production occurs in the pair of testes. Testes are present in scrotum outside the body. This is because optimum temperature for sperm production and survival is 34°C. Each testis contain about 800 seminiferous tubules. Seminiferous tubules walls contain two types of cells: Spermatogenic cells and Sustentacular cells. There are clusters of interstitial cells between seminiferous tubules. Thousands of sperms are produced in each second in seminiferous tubules. The process of sperm production is called spermatogenesis. Spermatogenic cells are also called spermatogonia (singl. spermatogomium) are present in the wall of seminiferous tubules. A Spermatogomium increases in size and becomes primary spermatocyte. Meiosis-I results in two secondary spermatocytes. After meiosis-II four non-motile haploid spermatids are formed. Spermatid undergo morphological changes and are transformed into motile sperms. A mature sperm consists of head whichcontain haploid nucleus and acrosome cap having acrosin, midpiece which have mitochondria and tail that contains microtubules for whip like movements. Sustentacular cells provide nourishment to sperms and maintain fluid environment while interstitial cells release male sex hormone testosterone.

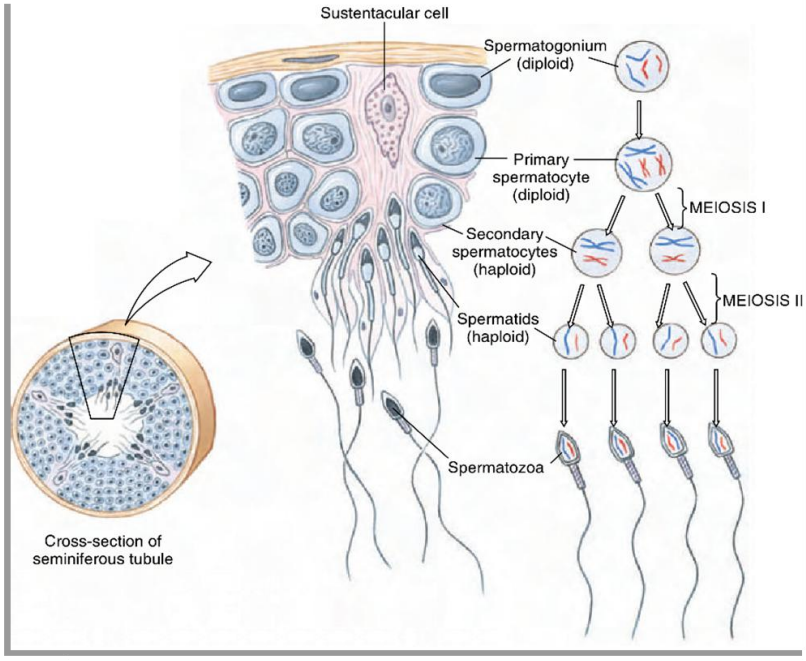


Fig. Spermatogenesis in seminiferous tubules of testis

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LESSON 147

Transport of sperms in Human

Spermatogenic cells produce sperms while ustentacular cells secrete fluid for nourishment of sperms.Sperms from seminiferous tubules comes into a network of called rete testis and then into epididymis where they are store for some time. Muscular contraction propel sperm forward. Sperms through duct deferens are transported to ampulla where stored till ejaculation.SeminalvesiclesSecrete water, fructose, prostaglandins and Vitamin C. This is energy source for sperm motility and helps to neutralize pH of female reproductive tract. Prostategland secretes water, enzymes, cholesterol, buffering salts and phospholipids. Bulbourethralglandssecrete clear alkaline fluid that lubricates urethra to facilitate ejaculation. Fluid that results from the combination of sperms and glandular secretions is called semen.Average human ejaculation contain 3 to 4 ml of semen having 300-400 million sperms.

LESSON 148

Hormonal Control of Male Reproductive Function

Male sex hormones are collectively called androgens (e.g., Testosterones). The hormones which travel from brain and pituitary to testis and ovaries are called gonadotropins (e.g., FSH and LH). Testosterone is secreted by interstitial cells it has role in development of testis, prostate and secondary sex characteristics.

Low level of testosterone is detected by hypothalamus. Hypothalamus release GnRH which effects the pituitary gland to release FSH and LH (ICSH). FSH stimulate spermatogenesis and LH stimulate the production of testosterone. Testosterone inhibits the secretion of LH while inhibin produced by sustentacular cells inhibits the secretion of FSH.

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LESSON 149

HUMAN FEMALE REPRODUCTIVE SYSTEM

Reproductive role of human female is more complex. The main functions of human female reproductive system are egg production, to nourish, carry and protect developing embryo. Female also nurse the young one after birth.

Human female reproductive system consists of ovaries, uterine tubes fimbriae and uterus, vagina, external genitalia and mammary glands. Ovaries produce eggs. They are also responsible for estrogen and progesterone production. Uterine tubes carry eggs from ovaries to uterus. Fimbriae are finger like projections which receive the egg into uterine tube. Uterus is the place where embryo is implanted for development. Vagina receives sperms during sexual activity. External genitalia are important in sexual arousal and have protective role while mammary glands are to feed the milk to newborn.

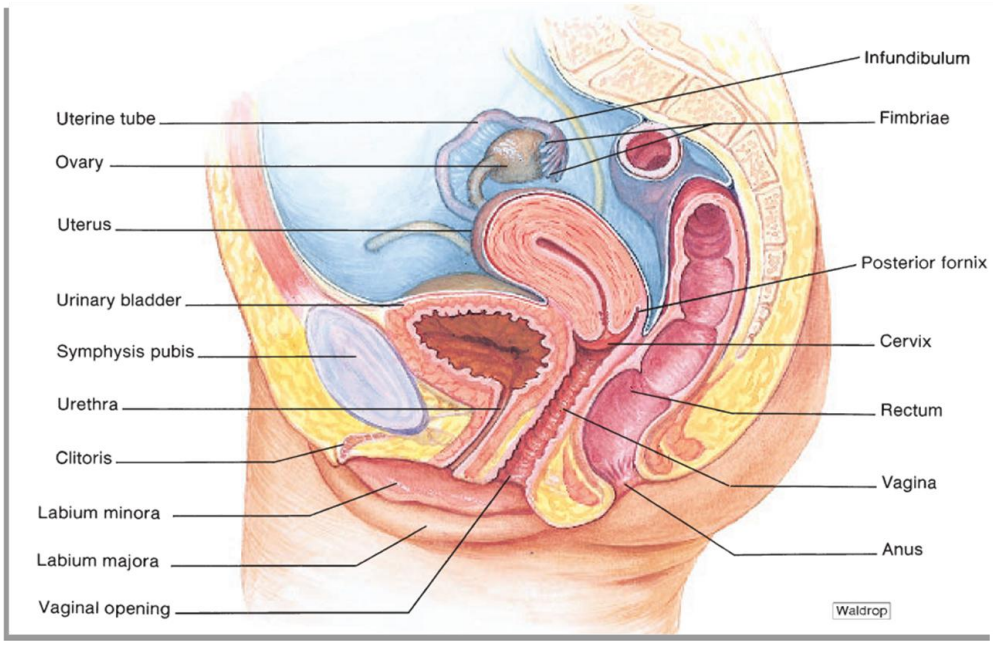


Fig. human female reproductive system

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LESSON 150

MENSTRUAL CYCLE IN HUMAN

It is the monthly cycle of changes in ovaries and preparation of uterus for implantation of embryo. The cycle begins at puberty and stops at the age of 45 to 55 years.

1. **The Follicular Phase:** Hypothalamus releases GnRH which stimulate the release of FSH and LH from anterior pituitary gland. FSH starts the development of oocyte in the ovarian follicles.

    i. **The Follicular Phase:** Developing follicle produces estrogen. Estrogen builds up and prepares uterus lining to become thicker (endometrium) for implantation of embryo.

    ii. **Ovulation:** Elevated level of estrogen increases the release of LH from pituitary gland and inhibits FSH. LH cause the mature follicle to enlarge rapidly and release secondary oocyte. This process is ovulation.

    iii. **The Luteal Phase:** Mature follicle after release of oocyte becomes corpus luteum. Corpus luteum secretes estrogen and progesterone. They complete the development of endometrium and maintain it for 10-14 days.

    iv. **Menstruation:** If the oocyte is not fertilized, corpus luteum disintegrates. Estrogen and progesterone production stops. As a result endometrium breaks and menstruation starts.

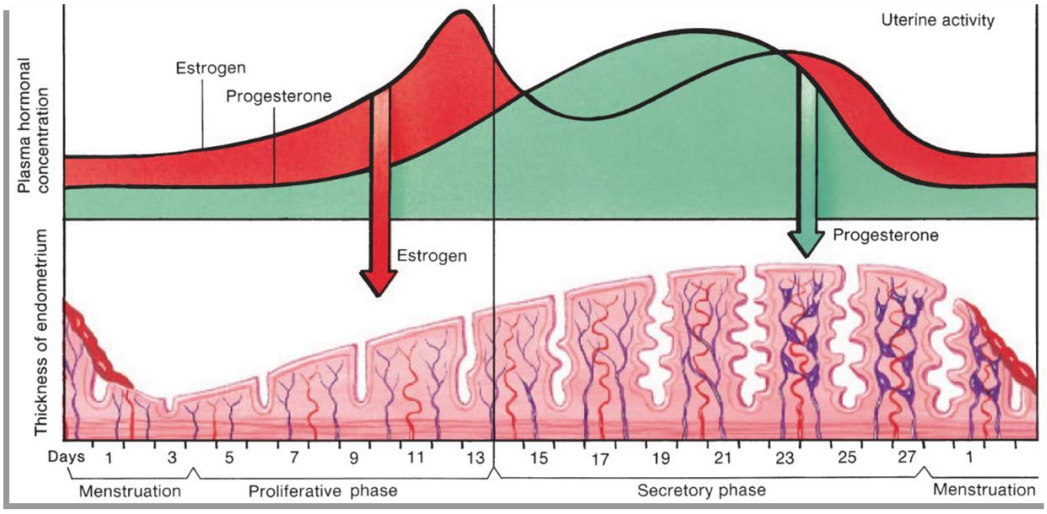


Fig. the human menstrual cycle

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LESSON 151

Follicle development in human

An ovary consists of an inner medulla and outer cortex.Medullahas blood vessels and nerves while cortex has developing follicles. During first five months of development many small follicles are formed in the ovary of female fetus. These are called primordial follicles. At the time of puberty during each menstrual cycle some primordial follicles began to develop further and become primary follicles which then converted to secondary follicles and graafian follicle. Many follicle start development in each cycle but only one completes all the stages. The follicle

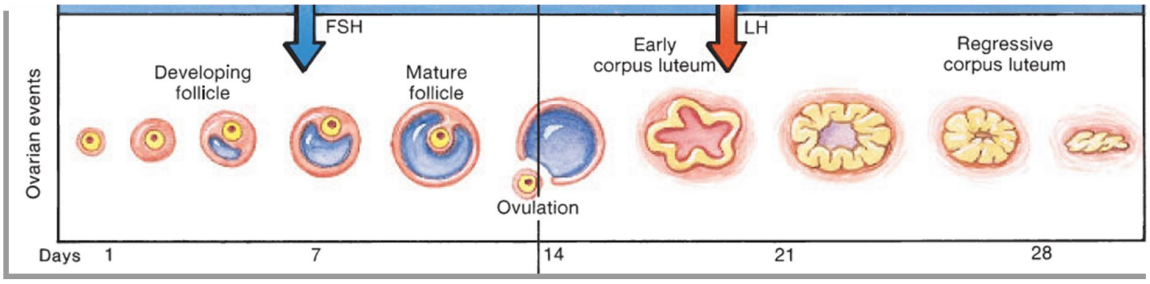


Fig. Different stages of follicle development

before release of egg is called graafian follicle. LH hormone causes the rupture of follicle to release egg. Graafian follicle after release of oocyte becomes corpus luteum which remain active for 10-14 days. If fertilization does not occur corpus luteum is converted to corpus albicans which ultimately disintegrates.

During each menstrual cycle several follicles start development but only one matures. Others disintegrates by the process atresia. In this process cells are disintegrated by apoptosis.

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LESSON 152

Production and Transport of Egg in Human

Egg production starts in female while still at the developmental stages. A definite number of eggs development takes place in specific structures called follicles. Egg development is halted at early stage of meiosis-I. At puberty during menstrual cycle meiosis-I is completed and meiosis-II starts which is completed after fertilization.

Egg is released from ovary by the process of ovulation. LH cause the release of oocyte from ovary. Ovary is encircled by finger like ciliated fimbriae. The ciliary action and muscular contraction cause the entry of oocyte into fallopian tube also called uterine tube. Fertilization takes place in upper one third portion of uterine tube. Oocyte completes meiosis-II after fertilization. Fertilization should take place within 24 hour of ovulation.

Egg moves due to ciliated cells and muscular contraction of uterine wall. Egg takes about 5 days to reach to uterus. If fertilized, embryo implanted into uterus otherwise egg is released out.

LESSON 153

Hormonal Control of Female Reproductive Function

Low level of estrogen stimulate hypothalamus to release gonadotropin-releasing hormone (GnRH). GnRH acts on pituitary gland to release follicle stimulating hormone (FSH) and luteinizing hormone (LH). Both of these hormones act on ovary. FSH start the development of follicle. 15-20 follicles start development but only one matures. Ovary also starts to produce estrogen. Estrogen causes the thickness of endometrium of uterus

Higher level of estrogen triggers LH and inhibit FSH form pituitary gland. LH from pituitary causes the ripening of ovarian follicle and release of oocyte from follicle. LH also convert collapsed follicle into corpus luteum after ovulation. Corpus luteum secretes estrogen and progesterone. Estrogen and progesterone have two roles: they maintain endometrium for completing development and also inhibit LH and FSH. If no fertilization occur, FSH and estrogen level drops and new cycle is started.

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LESSON 154

Historical Perspective of Embryology

Preformation theory: This theory says that gametes contain miniature version of all the element present in an adult. This means that every organ in a very small size is already present. Most biologist of 17th and 18th century believe in it.

Epigenesis theory gained popularity in mid-18th century. This theory says that embryo contain material from which embryo is gradually build. Much like a carpenter builds a building from nails, lumber and bricks.

Birth of modern embryology: Modern embryology emerged with the proposed germ layer theory. Embryo form 2-3 layers from which different body parts are formed.

Development Biology: It the study by which animals and plants grow. Developmental biologists use the tools of molecular biology to investigate the genetic control of development.

LESSON 155

Gametogenesis

Gametogenesis is the process of formation of gametes. It is of two main types: spermatogenesis and oogenesis.

Spermatogenesis: The process of sperms production in male gonads through meiosis is called spermatogenesis. Spermatogenesis takes place in testes. In human the entire process of spermatogenesis and its transportation takes about 3 months. However, it is different in different animals.

Testes contain diploid stem cells. Sperm mother cell is called spermatogomium. It enlarges in its size and become primary spermatocyte. Primary spermatocyte undergo meiotic cell division. After meiosis-I two secondary spermatocytes are formed. Then meiosis-II results in four haploid spermatids. Spermatids do not undergo any further cell division. They undergo morphological changes to produce sperms. Spermatids are non-motile while sperms are motile cells.

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LESSON 156

Oogenesis: The process of egg production in female gonads through meiosis is called oogenesis. Oogenesis takes place in ovaries. Egg formation and maturation takes place in follicles of ovary. Ovarian follicles contain oogonia. Oogonium increase its size and becomes primary oocyte which is a diploid cell with duplicated chromosome. Primary oocyte undergo meiotic cell division. After meiosis-I a secondary oocyte and a polar body is formed. Meiosis-II results in a haploid ootid and another polar body. Ootid undergo differentiation process to form ovum.

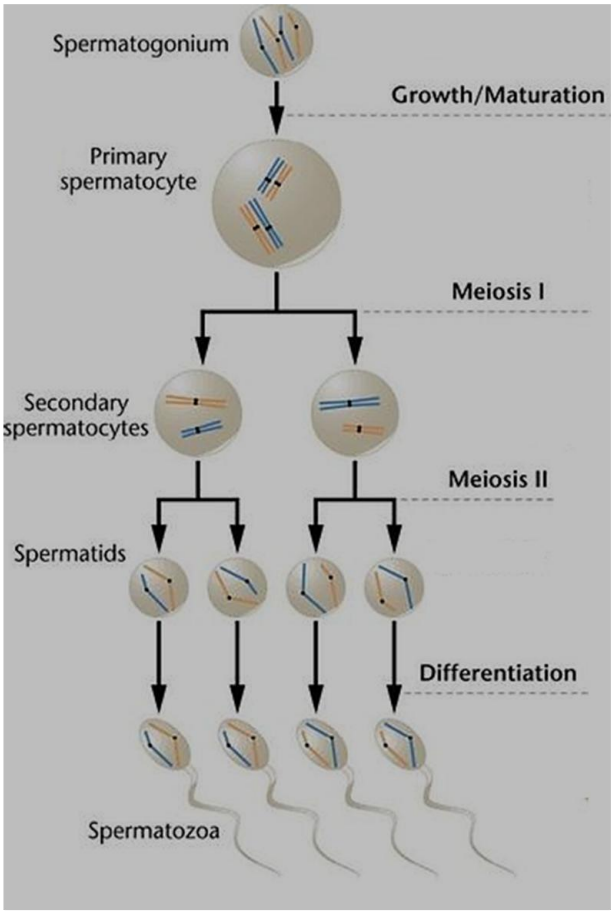


Fig. Spermatogenesis in animals

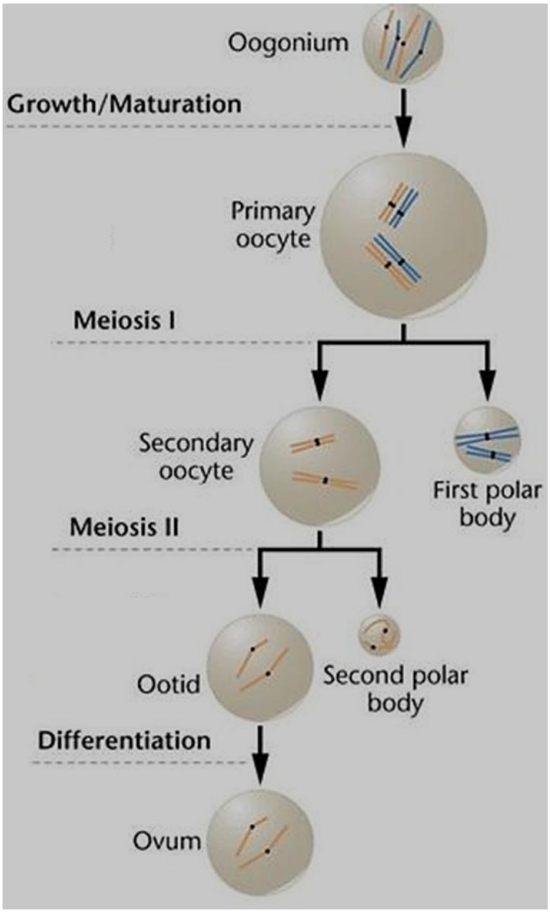


Fig. Oogenesis in animals

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LESSON 157

TYPES OF EGGS

Eggs may be classified on the basis of amount of yolk or distribution of yolk.

Classification on the basis of amount of yolk: On the basis of amount of yolk eggs may be alecithal, microlecithal, mesolecithal and macrolecithal. Eggs containing negligible amount of yolk are called alecithal egg. e.g., placental mammals. Eggs containing very little amount of yolk are called microlecithal eggs. e.g., sea urchin, amphioxus and marsupial mammals. Eggs with moderate amount of yolk are called mesolecithal eggs. e.g., lung fish and amphibians. While eggs with large amount of yolk are called macrolecithal eggs. e.g., reptiles and birds.

Classification on the basis of distribution of yolk: Eggs may be classified on the basis of distribution of yolk.Eggs with equally distributed yolk are called homolecithal eggs. e.g., sea urchin and mammals. Eggs with yolk in the center and cytoplasm in periphery are called centrolecithal eggs. e.g., insects. Eggs in which yolk is present at one end while cytoplasm at other end are called telolecithal eggs. e.g., frogs, reptiles and birds. The region of egg containing cytoplasm, its organelles and less amount of yolk is called animal pole while the region of egg containing more amount of yolk and less cytoplasm is called vegetal pole.

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LESSON 158

FERTILIZATION

Fertilization is the process in which egg and sperms fuse to form a diploid zygote which is the first representative of next generation. Fertilization is of two types: external fertilization and internal Fertilization. External fertilization takes place outside the body mainly in water, while internal fertilization takes place within the body of animal. Fertilization may be via copulation, cloacal connection or via spermatophore.

The structure surrounding plasma membrane of egg is called vitelline membrane. It is made up of fibrous proteins and receptors needed for sperm binding. The sperm contain an acrosomal head. The acrosome have specific enzymes called lysins which dissolve egg membrane during fertilization.

Many sperms approach to egg but only one successful. In some species only nucleus of sperm fuses with egg. In others, mitochondria, centriole and flagella of sperm may get entry into egg.

Egg Activation: It is a series of biochemical changes in egg which ensure the completion of fertilization and initiate embryonic development. Normally many sperms approach to egg but only one sperm should fertilize the egg. For this, second fertilization is stopped by changes in plasma membrane and outer region of cytoplasm.

LESSON 159

Membrane and corticular events: As sperm contacts, **microvilli** from egg plasma membrane wrap around the sperm. Contraction of microfilaments causes the sperm to draw into the egg. Within milliseconds of sperm penetration, the **chargeson plasma membrane** are changed. This makes plasma membrane unresponsive to other sperms. **Fertilization membrane** is formed by the discharge of granules of egg cortex. These granules are discharged between plasma membrane and vitelline layer. These granules loosen the contacts between plasma membrane and vitelline layer. Water molecules enter into the space and proteins of corticular granules thicken and strengthen the vitelline layer. Fertilization membrane formed within 1 to 2 minutes. After this, outer cytoplasm begins rotational and sliding movements. In amphibians, this rotation of cytoplasm results in the formation of **gray crescent** which is important in development.

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LESSON 160

Metabolic and nuclear events: After fertilization ionic changes occur in cytoplasm causing an increase in **intracellular pH**. This triggers the developmental process in zygote. The zygote starts the cleavage (cell division) process. For this, **DNA replication** starts as two copies of DNA are required for two daughter cells. Rate of **protein synthesis** is also boosted. This is because more and more proteins and enzymes are required for cell division and daughter cells. mRNA already present in the cell is activated to start protein synthesis. The mRNA in zygote used to protein synthesis is maternal in origin. Thus bulk of initial proteins is synthesized by maternal mRNA. This influence is called **maternal dominance**.

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LESSON161-162

CLEAVAGE AND ITS PATTERNS

Cleavage refers to cell divisions that occur during embryonic development. Initially all the cells of embryo divide at the same time this is called **synchrony**. As development proceeds, all cells don't divide simultaneously. This is called **asynchrony**. Eggs with evenly distributed yolk have cleavage pattern that result uniform size blastomeres. Egg with unevenly distributed yolk result in unequal size blastomeres. There are following important patterns of cleavage:

a. Holoblastic Cleavage

The pattern of cleavage in which egg is completely divided into two cells. Egg contain little or moderate amount of yolk. Cleavage in fertilized eggs of Amphibians, mammals, echinoderms, flatworms and nematodes.

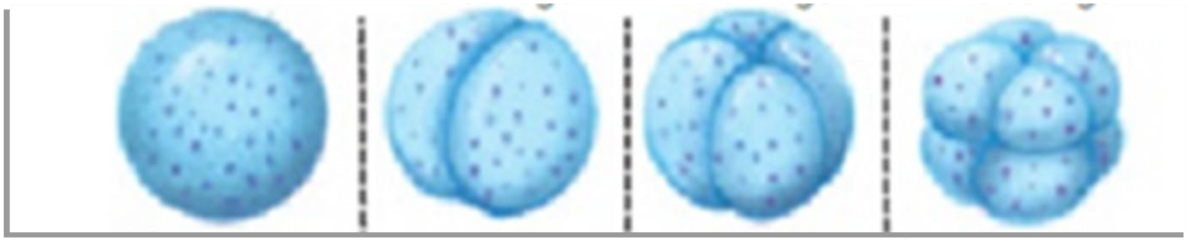


Fig. Holoblastic cleavage pattern

b. Meroblastic Cleavage

The pattern of cleavage in which egg is not completely divided into two cells. This is due to large amount of yolk. In these embryo development takes place around or on the top of yolk. e.g., cleavage in fertilized eggs of birds and reptiles.

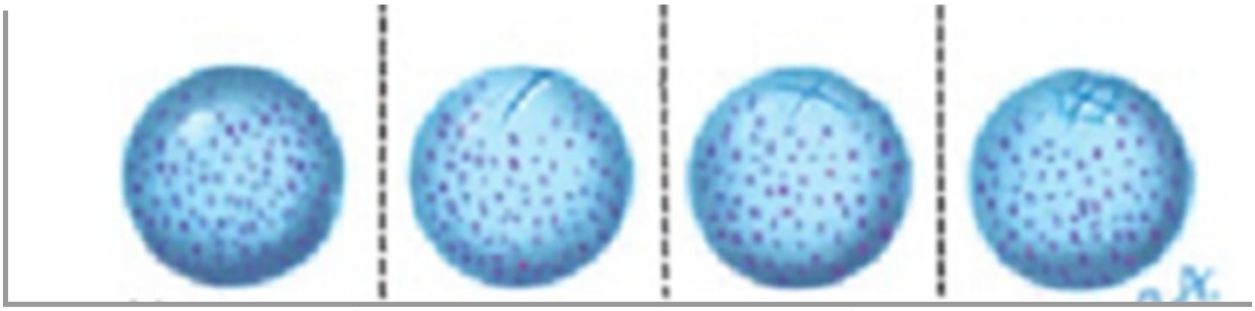


Fig. Meroblastic cleavage pattern

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LESSON 163

ECHINODERM EMBRYOLOGY

General Concepts

Echinoderms include starfish, sea cucumber, brittle star and sea urchin. Echinoderms are extensively studied group. This is because they are easily maintained in laboratory. Echinoderms lay eggs in water. In echinoderms fertilization is external and it takes place in water. Egg of echinoderm contain very little amount of yolk (microlecithal eggs).

LESSON 164

Cleavage: Cleavage is holoblastic and completely divides into two. Cleavage lines pass in such a way that cells lie exactly one above the other. This type of cleavage is called radial cleavage. Radial cleave is the characteristic of deuterostomes.

First two cleavage are from animal to vegetal pole (meridional) and perpendicular to each other. Third cleavage is equatorial and right angle to first two. It separates animal pole and vegetal pole. Fourth cleavage is medional. Initially division is equal, however, at sixteen cell stage division becomes unequal.

LESSON 165

Morulation and Blastulation: Cleavage results in a ball of cells called morula. Morula is a solid ball of cells without any cavity. At about 120 cell stage, a fluid filled cavity **blastocoel** is formed in the embryo. There is a single layer of cells around the blastocoel. Embryo at this stage is called blastula. At blastula stage embryo comes out of the fertilization membrane. At blastula stage cilia develop and embryo start swimming. At late blastula stage, some cells breakoff and position themselves within blastocoel. These cells are called primary mesenchyme and will form skeletal elements (spicules) of embryo. Late blastula is about 1000 cells of different size.

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LESSON 166-167

Gastrulation: Gastrulation is the process in which blastula is changed into gastrula. During this process primary germ layers are formed. Gastrulation begins at a point in vegetal half where cells start invagination. The point where invagination starts is blastopore. The invagination forms a new cavity in the embryo called **archenteron**. Archenteron gradually replaces blatocoel. Blastopore will form anus and archenteron is future gut of echinoderms. Some cells at the tip of developing archenteron get apart and form secondary mesenchyme cells. Secondary mesenchyme cells play a central role in pulling archenteron up to blastocoel wall. Initially embryo is round in shape. During gastrulation it begins to elongate and attains pyramidal shape (Prism) and ultimately form pluteuslarva. The larvae have a preferred direction of movement. Pluteus larva swims freely and feeds on smaller plants and animal. It has bilateral symmetry,

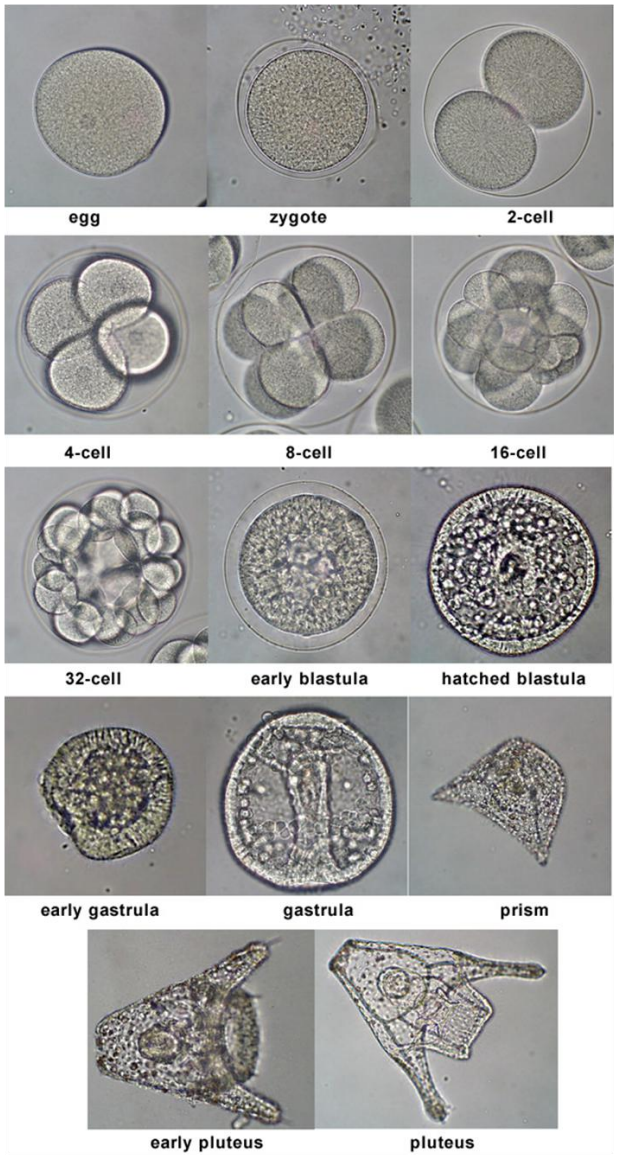


Fig. Echinoderm embryology

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LESSON 168

AMPHIBIAN EMBRYOLOGY

Most of the amphibians lay laid in water. Eggs are mesolecithal as they have moderate amount of yolk. The upper pigmented portion is called animal pole while the lower pole with large amount of yolk is called vegetal pole. Darkly pigmented animal pole on upper side camouflage from predators as it blends with bottom of water body. Light colour of vegetal pole blends with sky when viewed form below. Dark colour of animal pole also absorbs heat which promotes development. Fertilization takes place in water. Usually very little care by parents.

LESSON 169

Cleavage: Amphibians have mesolecithal eggs. Cleavage completely divide cell in two cells. Due to yolk at vegetal pole cleavage is slow. Size of cells at animal pole is smaller while larger at vegetal pole. First cleavage start from animal to vegetal pole. Cleavage at vegetal pole is slow due to large amount of yolk. It also divides gray crescent from middle. Second cleavage is perpendicular to the first and divides embryo into four. Third cleavage is equatorial and right angle to first two. Separate animal pole and vegetal pole. Fourth cleavage is meridional. Cleavage lines pass in such a way that cells lie exactly one above the other (radial cleavage). This radial cleavage is the characteristic of deuterostomes.

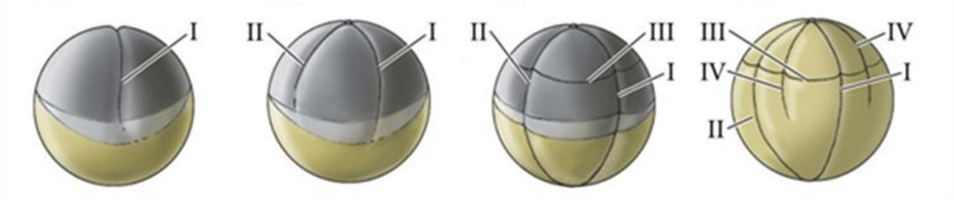


Fig. Cleavage in amphibians

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LESSON 170

Morulation: Cleavage results a ball of cells called morula. Morula is a solid ball of cells without any cavity. Cell towards animal pole are smaller while larger at vegetal pole due to presence of yolk.

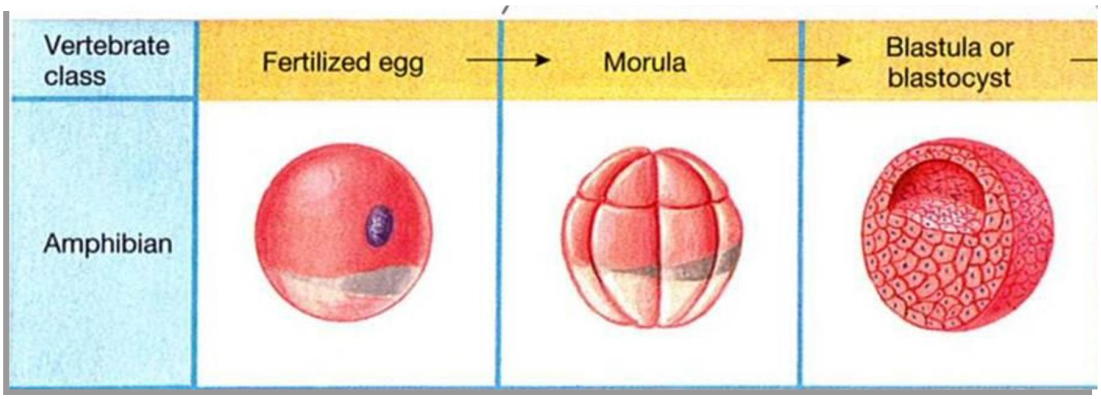


Fig. Morulation and blastulation

Blastulation: After morula formation, a fluid filled cavity **blastocoel** is formed in the embryo. Cell form multiple layers around the blastocoel. In amphibians blastocoel is formed towards animal pole due to yolk at vegetal pole

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LESSON171-172

Gastrulation: Gastrulation is the process during which cells arrange themselves into different layers called primary germ layers. Embryologist use dyes and carbon particles to mark different cells. Cells forming different structures are grouped on the surface of blastula. Later these migrate during gastrulation and form different structures. So these cells are named according to their future structure such as presumptive notochord, presumptive mesoderm and so on. Gastrulation begins with a slit like opening **blastopore** between gray crescent and ventral pole. Animal pole margin of blastopore is called its dorsal lip. Cells at the bottom of groove move anteriorly and groove spreads transversely. Cell at the surface begin to roll over the dorsal lip. This process is **involution**. Migrating cells spread across the slit like blastopore transversely and downward to the vegetal pole. Due to this migration one end of slit meets the other end. The ring like blastopore surrounds the yolk cells and they protrude out of the blastopore. This is called yolk plug. Later blastopore cells enclose yolk plug and blastopore is considered as closed.

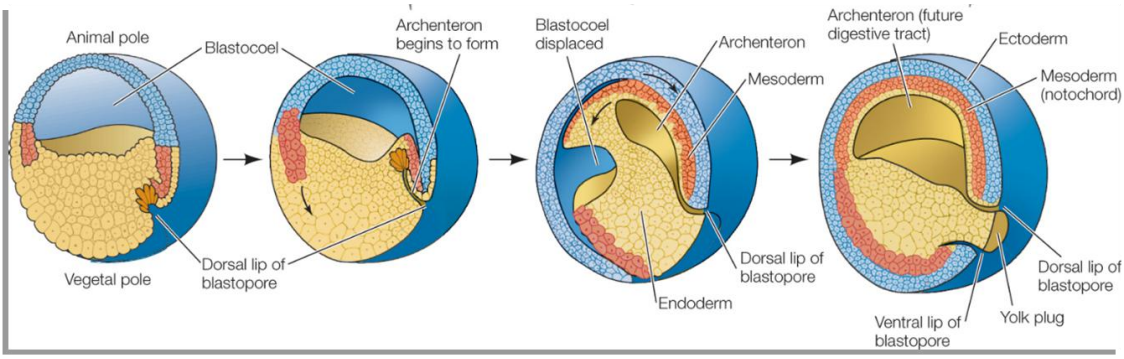


Fig. Gastrulation in amphibians.

Rolling over cells from blastopore form archenteron. Blastocoel becomes smaller and archenteron become larger. Archenteron is future gut and yolk cell make floor of gut. Some of the animal pole cell are rolling into the blastopore, they will form endoderm and mesoderm. The remaining animal pole cells are called ectoderm. These ectodermal cells migrate towards blastopore and spread over entire embryo. This is called **epiboly.** Some of the last cells rolling over the blastopore are presumptive notochord and presumptive mesoderm. Initially they form dorsal lining of archenteron near blastopore. Later they detach from endoderm and move between endoderm and ectoderm in the region of blastopore. This is called **chordamesoderm** which differentiates into notochord. Lateral to notochord mesodermal cell spread and thicken along sides of embryo. These thickenings are called **somites.** Somites appear as row of pumps from side view. Eventually somites spilt to form coelom

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For Final term exams

LESSON 173

Neurulation: The process of forming neural tube is called neurulation. An oval shape area on the dorsal surface of embryo is formed. Microfilament flatten and thicken this area. This region is neural plate. The edges of the neural plate roll up and over the midline of neural plate. These raised edges in longitudinal direction are called neural folds. The raised neural folds meet with each other longitudinally over the midline of neural plate. Now neural plate is converted into a tubular structure called neural tube. The portion of neural tube that is close at the end will form brain

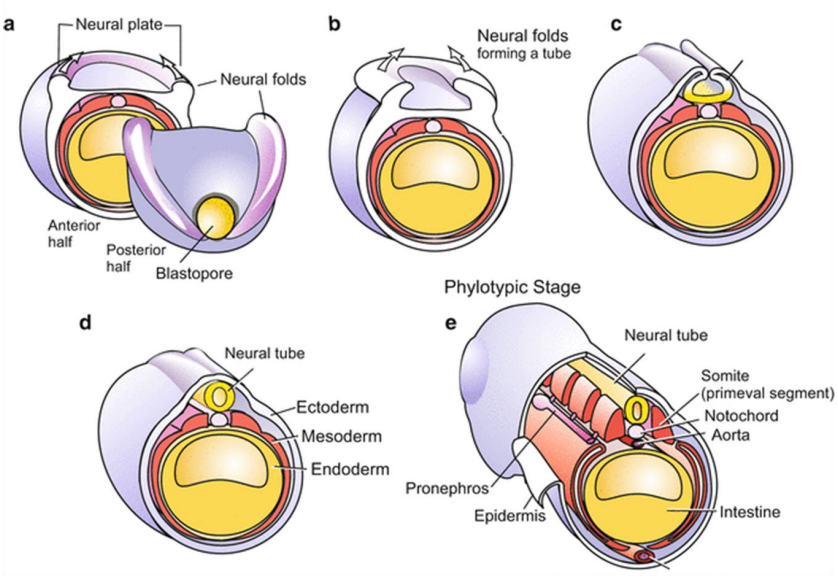


Fig. Neurulation in amphibians

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LESSON 174

AVIAN EMBRYOLOGY

Chicken development is used as a model for the development of birds and reptiles. Birds egg is macrolecithal and contains large amount yolk. Due to large amount of yolk egg nucleus is present at one side, so they are called as telolecithal eggs. Yolk is also protected by albumin $CaCO_3$ shell. In birds fertilization is internal and it takes place in upper portion of oviduct. Development starts right after fertilization while egg is still in oviduct. All developmental stage are completed within shelled egg. In chicken development is completed in approximately three weeks.

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LESSON 175

Structure of Avian Egg

Membranes and fluids are collected around the egg while moving through oviduct. Water and albumin is deposited around the egg. This protect from desiccation and mechanical damage. Albumin is used for protection and a source of nutrition. Two denser strands of albumin called chalaza are attached to the egg shell. They suspend egg in the center of albumen. Shell gland secrete $CaCO_3$ which deposit around the egg. It is also impregnated with proteins. Thousands of tiny pores (40-50µm) are for exchange of gases. Inside the shell two membranes are present. At the round end of egg an air pocket is present between two membranes. Air pocket enlarges with time. Near the time of hatching the chick breaks air pocket, the air inflates its lungs and the chick start breathing.

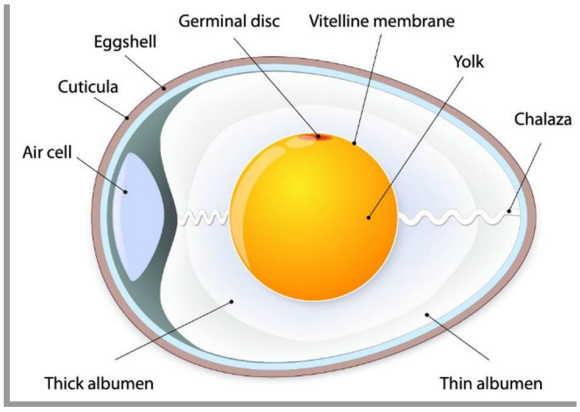


Fig. Structure of chicken egg

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LESSON 176

Cleavage: Due to large amount of yolk cleavage do not divide cell in two. There is no division in the large amount of yolk. All the divisions are at one end of egg forming a disc shape structure. Such a cleavage is called discoidal cleavage. Cleavage is radial as cleavage lines pass in such a way that cells lie exactly one above the other. Radial cleave is the characteristic of deuterostomes. First two cleavage are vertical and perpendicular to each other. Third cleavage is equatorial and right angle to first two. Fourth cleavage is vertical.

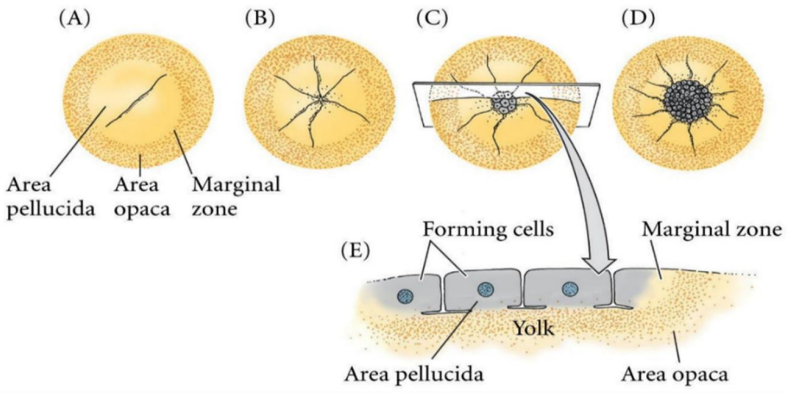


Fig. discoidal cleavage in chick embryo

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LESSON 177

Blastulation: The cells in the animal pole dived. When the cell are approximately 60,000, they form a blastoderm. The yolk below blastoderm is used which forms a fluid filled cavity called blastocoel. Blastoderm splits into two layers: Outer layer is called epiblast while the inner layer is hypoblast.

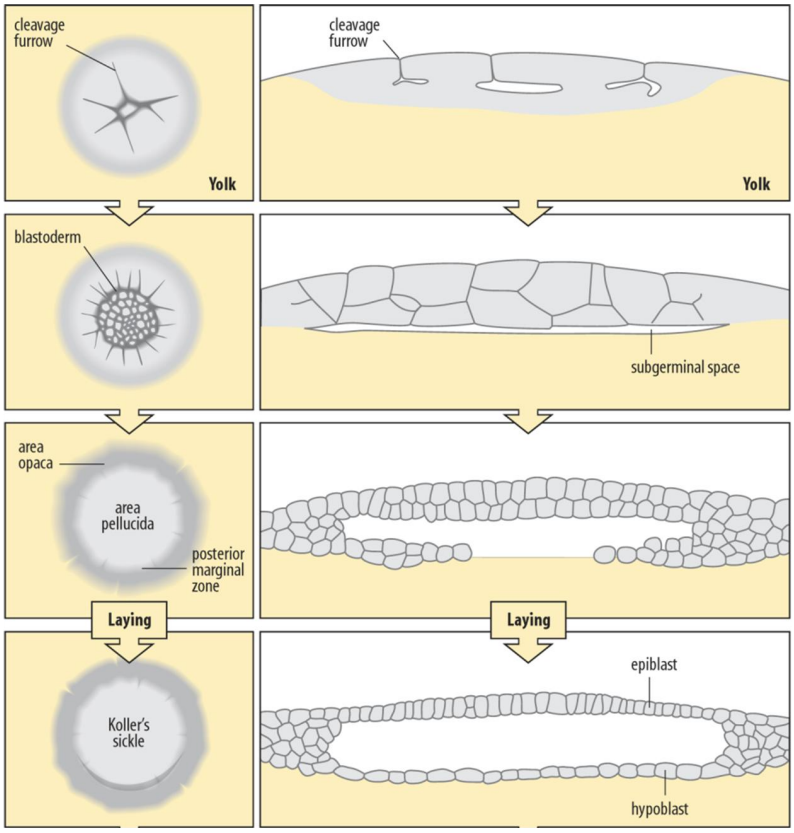


Fig. Blastulation in chick embryo

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LESSON 178-179

Gastrulation: The dorsal cells of blastoderms form a medial linear invagination called primitive streak. Primitive streak gradually extends anteriorly. It defines the axis of embryo. It also separate right and left side. The anterior margin of primitive streak there is a regional thickening called Henson's node.

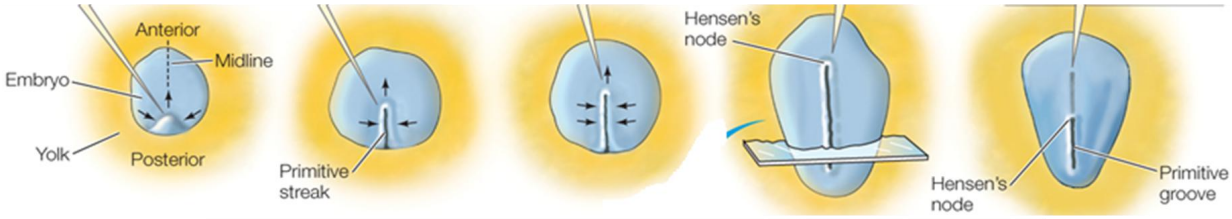


Fig. primitive streak and Hensen's node in chick embryo

Henson's node contains funnel shaped depression through which cell enter between epiblast and hypoblast. Henson's node is functionally equivalent to blastopore of amphibians. First cells migrating through Henson's node move anteriorly and ventrally to form pharyngeal endoderm of foregut. Hypoblast cell are moved anteriorly. Next migrating cells move anteriorly, but not far away. They remain between endoderm and epiblast to form prechordal plate mesoderm. They form head process. After that the migrating cells form chordamesoderm which forms notochord. Cells that ingress through middle part of streak give rise to middle part of somites and to heart and kidney. Cell which ingress through posterior end give rise to lateral plate and extra embryonic mesoderm.

After migration, the cell remaining outside form ectoderm. Ectodermal cells close to streak form neural tube and the cells farther from streak will form epidermis. Henson's node attains a more posterior position due to regression of primitive streak. Now the epiblast cells are all ectodermal cells. The ectodermal cell migrate and enclose the yolk. This process is called epiboly. Embryo lift it off the yolk. Only a connection is retained between embryo and yolk. Blood vessels develop in yolk stalk to carry nutrients from yolk.

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LESSON 180

MAMMALIAN EMBRYOLOGY

Class mammalia is divided into three subclasses. Prototheria are egg laying mammals. Their eggs have large amount of yolk. Marsupial mammals are pouched mammals. Their eggs have verty little amount of yolk (microlecithal eggs). Placental mammals have amost no amount of yolk (alecithal eggs). Placental mammals egg is smallest in animals kingdom. Mammals have internal fertilization. Placental and marsupial mammals have internal development and both give birth to young ones.

Marsupial mammals have premature birth. They keep the newborn in marsupium and nourish with their nipples. Placental mammals have long gestation period. Fetus is fed through placenta.

LESSON 181

Cleavage and Blastulation: Mammalian cleavage is slowest in animal kingdom. Blastomeres have unique orientation. There is rotational cleavage. First cleavage is meridional while in second cleavage one is meridional and other is horizontal. Cell division is asynchronous in the very beginning. Cell are not 2-4-8, but may be odd number. Cell switche from maternal to zygote control at 2 cell stage for proteins synthesis. Through 8 cell stage, loose arrangement with plenty of space. After third cleavage the cells tightly huddle together to form a compact ball. Eight cells divide to produce 16 cell **morula**. A small group of internal cells is called **inner cell mass** (ICM) while the outer cells form **trophoblast.** ICM form embryo while trophoblast forms chorion for placenta formation. 8 cells stage blastomere can form both ICM and trophoblast, they are called **totipotent.** ICM cells can form any type of embryo cells, they are called **pluripotent**. Trophoblast cells of morula secrete fluid that create blastocoel. As blastocoel expands ICM and trophoblast take their positions. At this stage blastula is called **blastocyst.**

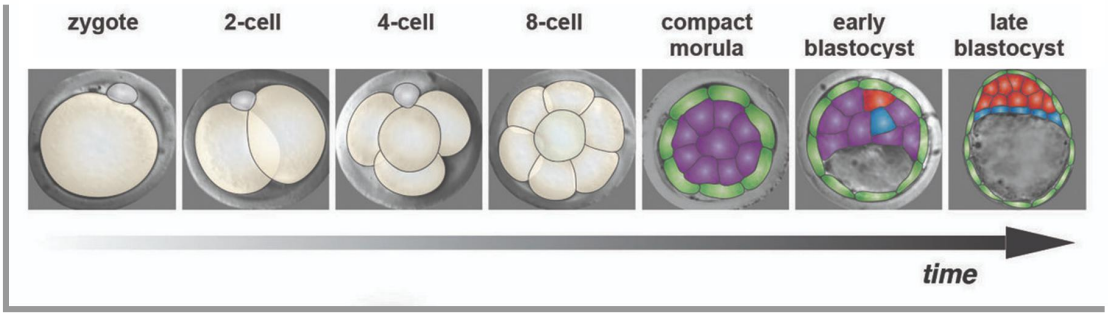


Fig. Cleavage and blastulation in mammals

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LESSON182

GASTRULATION: Both mammals and birds originate from reptile. Despite difference in amount of yolk both have striking similarity in development. First segregation of cells within inner cell mass to form hypoblast and hypoblast. Both epiblast and hypoblast form a structure **bilaminar germ disc.** Hypoblast cells form extraembryonic endoderm which form yolk sac. Epiblast form embryonic endoderm and amnion cavity. Epiblast cells move through primitive streak for form endoderm and mesoderm. The cell anterior and lateral to primitive streak form ectoderm.

LESSON 183

HORMONAL REGULATION IN PREGNANT FEMALE

The cells developing embryo and placenta release **human chorionic gonadotropin** (hCG). hCG maintains the corpus luteum so it may not disintegrate. High level of hCG is used as immunological test for pregnancy by 2 weeks. **Progesterone** is released by corpus luteum. Progesterone maintains the endometrium lining to retain pregnancy. **Prolactin** is secreted by pituitary gland. It is produced in pregnant female to start the process of milk production and eject milk after childbirth. **Oxytocin** is also produced by pituitary gland. It caused contraction of smooth muscles to start labour and expulsion of young one. It also stimulate nipples to release milk. **Prostaglandins** secretion increase during labour. They also cause smooth muscle contraction.

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LESSON 184

PRENATAL DEVELOPMENT IN HUMAN

In human the development can be divided into two phases: prenatal (before birth) and postnatal (after birth) phases. After two weeks of fertilization till the end of 8th week it is called embryo and from 9th week till birth is called fetus. After the birth it is called newborn or baby. The total gestation period in human is 280 days. This duration is divided into 3 months periods called trimesters.

First Trimester starts from fertilization. Zygote divides to become morula. By 4th day it is converted into 50- 120 cell blastocyst. As embryo reaches into uterus, it implants itself onto uterine wall. Trophoblast invade the endometrium. Gastrulation is similar to birds and reptiles. Once gastrulation completed different organ formation starts called organogenesis. By middle of first trimester all body organ begin to develop and embryo also grows in size.

LESSON 185

Second Trimester: It ranges from 14 to 27 week. Almost all major and vital organs have been completed during this trimester. Growth is magnificent. Heart beat can be heard by the end of fourth month (18 Week). Eyes become open and bones ossification started.

LESSON 186

Third Trimester: It start from 7th month till birth. Fetus has potential to survive if born premature in this trimester. Hair and nail grow in this trimester and the fetus weighs almost double.

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LESSON 187

THE PLACENTA

Placenta is s temporary organ which connects developing fetus to mother via umbilical cord. Human have placenta formed form chorion and allantois. The tiny chorionic villi sent out from blastocyst form placenta . Placenta formation starts from day 6 and is completed by 1st trimester. Embryonic blood vessels of placenta do not merge with maternal blood vessels. Placenta is about 22cm in length. Placenta is used for nutrient uptake, thermoregulation, waste elimination, gaseous exchange, fight against internal infection and placental hormone production. Placenta is connected to fetus through umbilical cord which contains two fetal umbilical arteries and one fetal umbilical vein. They spiral about each other in umbilical cord. Average size of umbilical cord is 55cm with 1-2cm diameter.

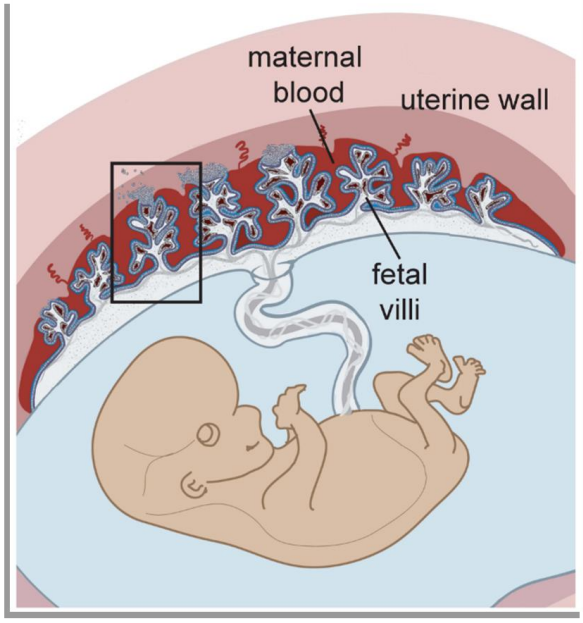


Fig. placenta and umbilical in human

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LESSON 188

BIRTH IN HUMAN

Gestation is the total time period which embryo or fetus spends within body of mother. In human it is 280 days or 40 weeks. The process of birth is called parturition. During parturition, mother's uterine muscles begin to contract and cervix begins to dilate. Mother ovaries and placenta produce relaxin hormone. Relaxin causes mother pelvic bone to slightly separate, so the baby may pass from birth canal. During birth baby pituitary gland secretes Adrenocorticotropic hormone (ACTH). ACTH stimulates adrenal glands to secrete prostaglandins which work along with oxytocin. Stretch in uterine wall stimulate pituitary to secrete oxytocin hormone in mother. Oxytocin and prostaglandins cause powerful muscular contractions. Usually within an hour of muscular contraction baby is expelled out from uterus. After baby comes out the process of contraction continue to expel placenta, the process is called afterbirth.

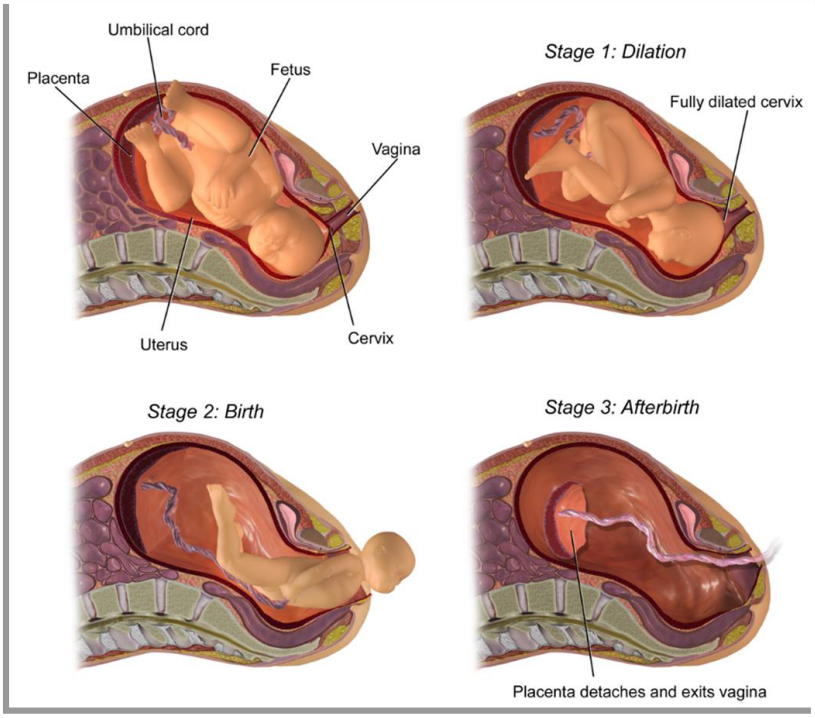


Fig. Stages of birth in human

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LESSON 189

MULTIPLE BIRTHS

The organisms which give birth to only one offspring at a time are called **uniparous**. *e.g.,* human, cows, bufallos. Those animals which give birth to more than one young ones are called **multiparous.***e.g.,* mice, rat, dogs, cats. Usually the young ones are both male and female. However armadillo give birth to four young ones at same time, all males or females.

Monozygotic Twins are from single zygote and are identical twins. They are strikingly alike and of same sex. About 33% have separate placenta. Majority have separate amniotic sac. Small proportion have one placenta and one amniotic sac which indicate separation occur after 9 day of pregnancy. There is conjoining risk, the condition is Siamese twining.

Dizygotic Twins are produced from two zygotes. They are called as fraternal twins. They do not have striking similarities and may be of same or different sex. They have separate placenta and separate amniotic sac.

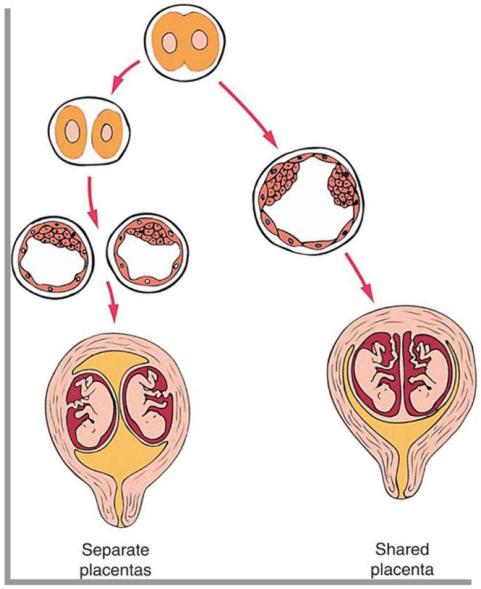


Fig. Monozygotic twins in human

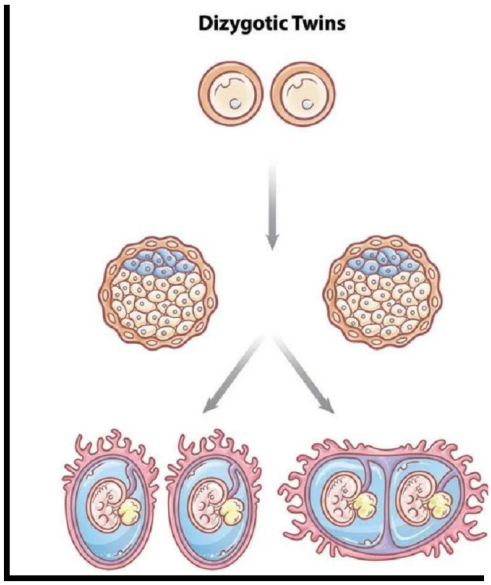


Fig. Dizygotic twins in human

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LESSON 190

MILK PRODUCTION AND LACTATION

Mammary glands are unique feature of mammals. They are evolved from sweat glands in the skin. The process of milk production by the mammary glands is called lactation. During pregnancy breasts enlarges due to hormone prolactin. Estrogen and progesterone inhibit the secretion of milk before birth. After birth both decrease and milk production starts. Newborn stimulate pituitary to release oxytocin and prolactin. Prolactin produce milk while oxytocin triggers its release from mammary glands. The higher protein fluid in breast at the time of birth is called **colostrum.** It contains maternal antibodies to boost immune system. Colostrum works as laxative to remove meconium (stool of infant). Mother's breast don't release milk for 1-3 days.

LESSON 191

EXTRAEMBRYONIC MEMBRANES

Extraembryonic membranes are yolk sac, amnion, chorion and allantois. Reptiles, birds and mammals embryos have amnion, chorion and allantois are called amniotes whereas eggs of fishes and amphibians having amnion, chorion and allantois are called anamniotes. Amnion encloses embryo in a fluid filled sac. Protects against shock and desiccation. Chorion is near to shell, mechanical support, highly vascular, gaseous exchange. Allantois is for storage and excretion of toxic wastes, also for gaseous exchange. Yolk sac is involved in distribution of nutrients to developing embryo.

Ectoderm and mesoderm on both side of embryo lift off the yolk and grow dorsally, they fuse to form chorion (outer side) and amnion (inner side). A ventral outgrowth of gut tract give rise to allantois. It gradually increase in its size along with development. Yolk sac develops from proliferation of endoderm and mesoderm around yolk.

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LESSON 192

PRIMARY GERM LAYERS

During gastrulation three germ layers are formed. These are ectoderm, mesoderm and endoderm. These germ layers give rise to different body parts.

Ectoderm: It is the outer most layer of cells formed during gastrulation. Some of the ectodermal cells aligned themselves into a columnar pattern to form **neural plate** which is converted into neural tube. Neural tube will form central nervous system while cells of neural crest will form peripheral nervous system. After neurulation all ectodermal cells form epidermis. Epidermis also give rise to hair, sebaceous glands, sweat glands, nails, claws and hoofs.

LESSON 193

Endoderm is the inner most primary germ layer formed during gastrulation. The entire gastrointestinal tract except mouth, pharynx and anus develop from endoderm. Endoderm also give rise to the lining of all glands which are part of digestive system e.g., liver, gall bladder and pancreas. Endoderm also give rise to respiratory tract, thyroid, parathyroid, thymus, pancreas, auditory tube, tympanic cavity and urinary system.

LESSON 194

Mesoderm is one of the primary germ layer formed during gastrulation. It is present between ectoderm and endoderm. Mesoderm give rise to skeletal system, urinogenital structures, circulatory system, muscular tissues, skin, tendons, ligaments and adipose tissues. Coelom is also formed by out pouching or splitting of mesoderm.

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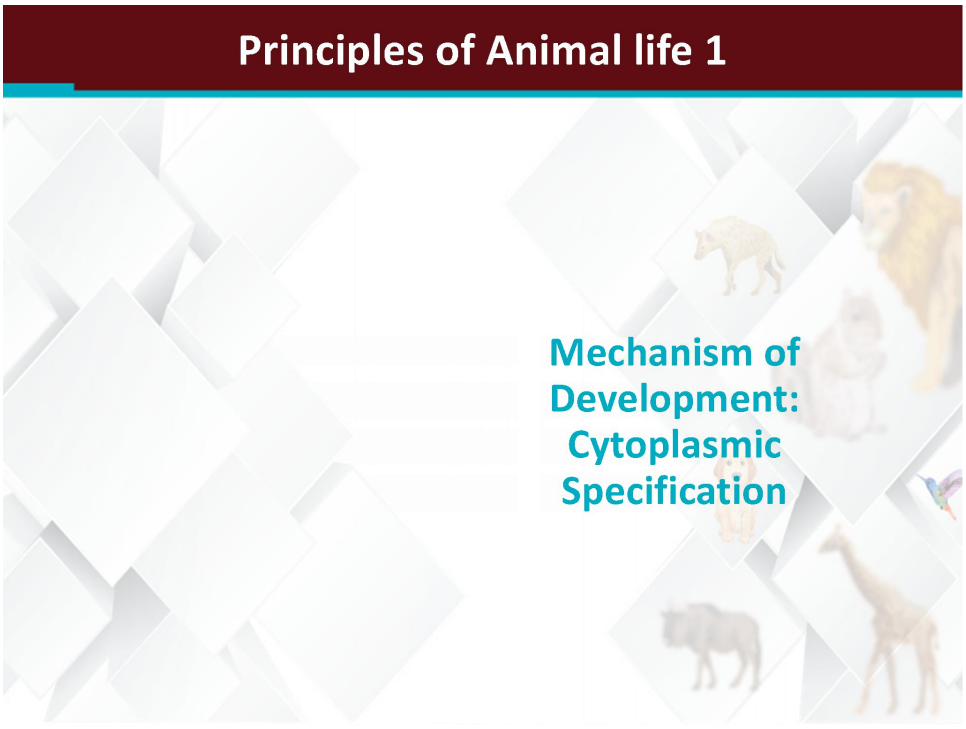
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MECHANISM OF DEVELOPMENT

LESSON 195

Nuclear equivalence

In nineteenth century, scientist were working that how developing embryo develop multiple types of cells? Roux-Weismann gave the hypothesis that genome is broken down into smaller and smaller fragment. This is the reason that different cells give rise to different types of cells. In 1892, Hans DrieschDiscovered that if two cell stage sea urchin embryo is mechanically shaken to get cells apart, both develop into normal larvae. This shows that both cells contain same genetic material. Hans Spemann placed a minute ligature of hair around salamander zygote. Nucleus containing portion remain dividing while the other didn't. Later at 16 cell stage, one nucleus move to anucleated portion and it also started cell division. When separated both developed into normal embryos. Hans Spemann further concluded that both halves developed normally only if both receive gray crescent. This means cytoplasmic contents paly their role in formation of different types of cells.



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LESSON 197

Embryonic Induction

The capacity of some cells to evoke developmental process in others is called embryonic development. It is wide spread phenomenon in development. Spemann and mangold first reported embryonic induction. When the dorsal lip of blastopore was transplanted to lateral or ventral side of other embryo, it developed notochord, somite, ectoderm of host and formed the whole system formed. Dorsal lip of blastopore can induce the development of whole embryo, it is called as primary organizer. The phenomenon of induction by blastopore is called primary induction. Formation of other type of cell by inductive process is termed as secondary induction.

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LESSON 198-199

PROTOSTOME VS DEUTEROSTOME

Protostomes belong to group protostomia of animal kingdom. They are part of grade bilateria along with deuterostomes. Platyhelminthes, rotifera, nematoda, annelida, arthropoda and Mollusca are the member phyla of protostomes.

Deuterstomes belong to group deuterostomia of animal kingdom. They are part of grade bilateria along with protostomes. Echinodermata, hemichordate and chordata are the member phyla of deuterostomes.

a. Both differ in cleavage pattern. In protostomes cleavage is spiral while in deuterostomes cleavage is radial.
b. In protostomes cleavage is determinate as the fate of blastomeres is established at very early stage *i.e,* 2 or 4 cell stage. In deuterostomes cleavage is indeterminateas the fate of blastomeres is established at later stages of development.
c. In protostomes blastopore develops into mouth while in deuterostome blastopore develops into anus.
d. In protostomes coelom is formed by splitting of mesoderm. This pattern is called schizocoelous while in deuterostome coelom is formed by out pouching of primitive gut (archenteron). This pattern is called enterocoelous.
e. Development of many protostome include a top shape larva called trochophore larva while some deuterostome have a Kidney-bean shaped larva called dipleurula.

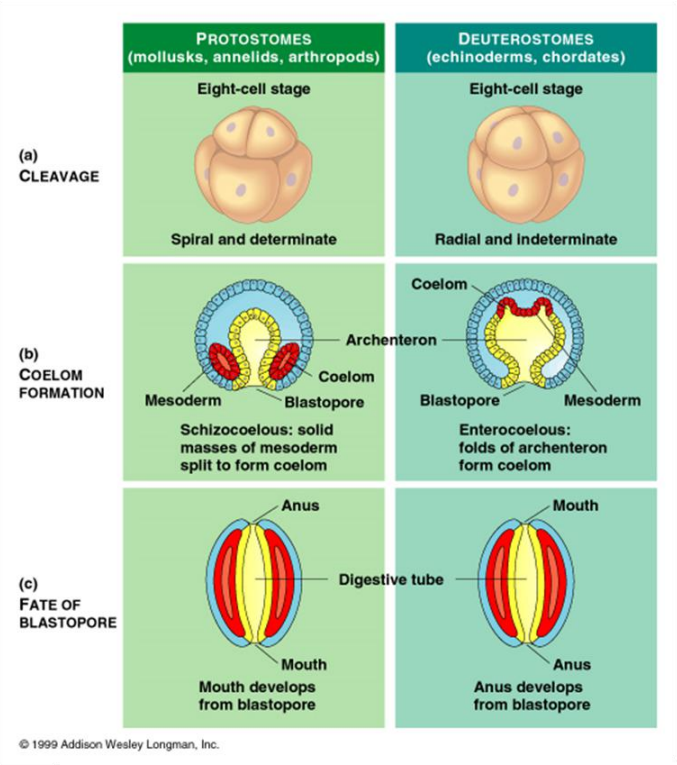


Fig. some major differences between protostome and deuterostome

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LESSON 200

METAMORPHOSIS

Metamorphosis is an abrupt change in structure and function from immature stage to adult. It takes place in insects, mollusks, crustaceans, cnidarians, echinoderms, tunicates, fishes and amphibians. During metamorphosis animals undergo a series of molts to grow, and each stage between molts is called an instar e.g., mosquito larvae have four instars and butterfly larvae have five instars.

Metamorphosis is beneficial for animals because animal uses different types of food at different stages. This reduces the competition between juvenile and adult stages e.g., tadpole of frog is herbivore while adult is insectivore.

The disadvantages of metamorphosis include short life of adult while metamorphosis stages are pretty long. As larva and adult do not share same food, this may be disadvantageous in food poor environment.

Hormonal Control of Metamorphosis: Metamorphosis is controlled by thyroid hormones in vertebrates. In insects prothoracicotropic hormone from insect brain stimulates ecdysone hormone released by prothoracic gland. At larval stage juvenile hormone is also released this results in next larval instar. At final stage of larva, juvenile hormone is not released and ecdysone converts larva into pupa.

Metamorphosis is of two types: complete metamorphosis (holometabolous) and incomplete (hemimetabolous) metamorphosis.

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LESSON 201

Holometamorphosis(Complete metamorphosis) occurs in about 88% insects. It is the type of metamorphosis which separates the physiological process of growth from differentiation and sexual stages. There are four stages i.e., egg, larva, pupa and adult stage. Each process is completed without intervention of other stage.

Early embryonic development completed at egg stage. Larval stage is motile and worm like form. Larva actively feed and grow in size. It never possess wings and have simple eyes. Larva undergo metamorphosis to become pupa. It is quiescent, non-feeding and mostly very little motile. During pupa stage, there are drastic changes in morphology and physiology. Adults have wings and reproductive organs. Most insects adult grow very little. Some do not feed and entirely focus on mating and reproduction.

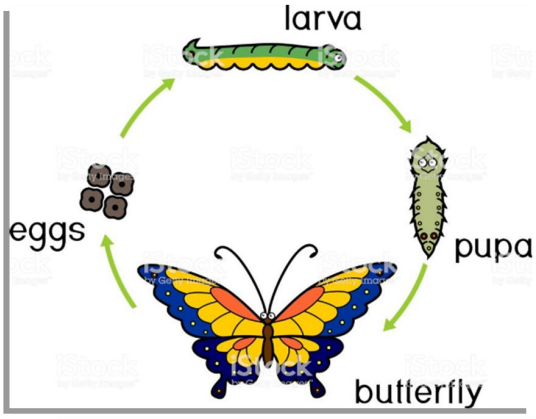


Fig. Complete metamorphosis

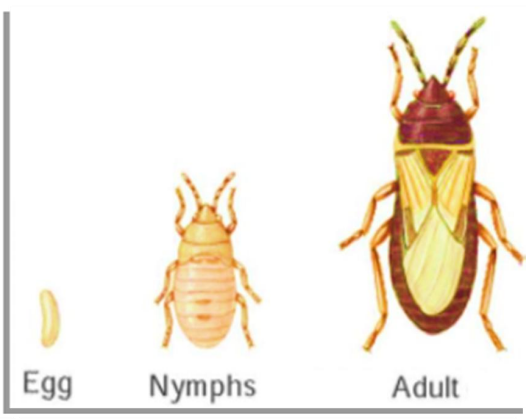


Fig. Incomplete metamorphosis

LESSON 202

Hemimetabolous metamorphosis (Incomplete metamorphosis) is the mode of development of certain insects like aphid, grasshopper and cockroach. They have three distinct stages i.e., egg, nymph and adult.

Initial stages of development are completed at egg stage. Nymph is an immature form hatched out from the eggs. It lacks wings and have underdeveloped reproductive organs. They adult have wings and reproductively mature.

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Principles of Animal life 1



Principles of Animal life 1



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Principles of Animal life 1



Principles of Animal life 1



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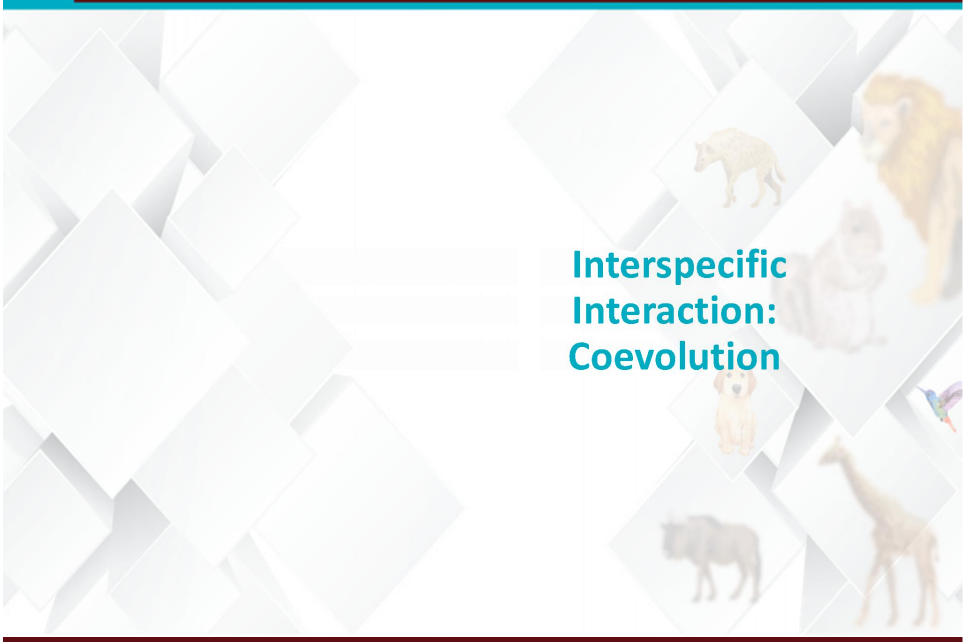
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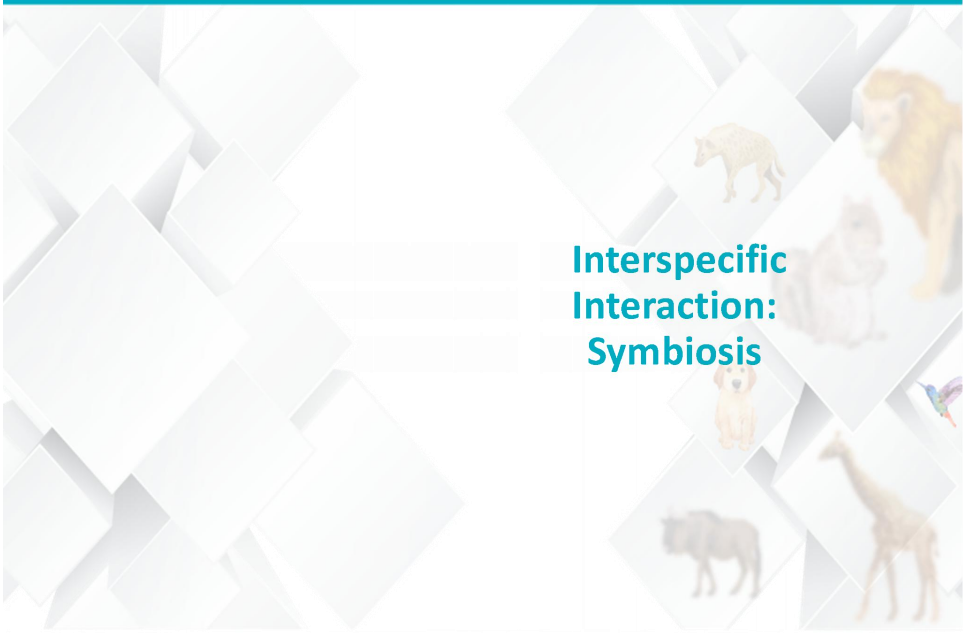
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Principles of Animal life 1



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LESSON 220

a. Predation

It is type of interaction in which on animal kills and eat other animals. The animal which eats is called predator and the animal being eaten is called prey. Interaction between predator and prey are complex. A large population of prey supports more number of predators. A decreased population of prey results in a decreased population of predators. Predators also effect the distribution of prey. For example, the lake trout of North America has never moved to freshwater of Siberian freshwater system. This is probably due to predation by Pacific sea lamprey.

LESSON 221

i) Commensalism

It is the symbiotic relationship between two organisms in which one member benefited while other is neither helped nor harmed. It is a small fish with disk on head. By these disks they get attached with large sharks and whales. When shark or whale feed they eat the scrapes of food. Similarly, tree frogs live in green leaves of tree. They feed on insects and are protected from predators in green leaves of plants. A commensal do not harm its host. However, sometimes environmental factor change and a commensal becomes parasite. e.g., *Entamoebagingivalis*lives in mouth of human may cause teeth and gum decay occasionally.

LESSON223

a. Camouflage

It is the phenomenon in which the colour pattern of animal helps to hide the animal from others. There are two major types of camouflage: cryptic coloration and countershading. In **cryptic coloration** camouflage animals takes pattern according to surroundings. Some prey animals are camouflage from predators. e.g., arctic hare white fur coat in winter takes pattern of snow and brown in spring takes pattern of soil. Similarly predators use camouflage to avoid detection by prey. For example, Asian tiger's rufous-red coat with black strips help it to undetected during stalking. **Countershading** camouflage is common in eggs of frog and toad. Eggs are darkly pigmented on top while light pigmented on the bottom. From top dark top mixes with dark bottom. From bottom light colour blends with light air water interface.

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LESSON224

a. Aposematic Coloration and Mimicry

Phenomenon in which animals protect themselves by advertising themselves dangerous or distasteful is called **aposematic coloration**. They actually warn the predators to not attack on them. Dandrobatid frogs of central and South America have poisonous secretion and are brightly colored. Their bright colour is a warning to the predators.

Mimicry is the phenomenon in which one species gains resemblance with other species and gets protection from predators. This resemblance may be with one or more than one species. In **batesian mimicry,** one species resembles with other species and protected by aposematic coloration. Flies belonging to family Syrphidae resemble with bees or wasps, thus protected. Here flies are mimic and bee or wasps are model. **Mulerian mimicry** both mimic and model both are distasteful. If predator eats any one of these, it will not prey again any one of them.

LESSON225

COMMUNITY STRUCTURE

Communities are not the random mixture of populations but they are organized entities. Some members of community have greater importance in organizing community. Species that is responsible for determining community characteristics is called **dominant species**. e.g., a stream having a large population of rainbow trout that will determine the population of certain invertebrates. Those species on which several other species are dependent are called **keystone species**. Removal of keystone species will cause the death of dependents. Cyclic activities also characterize the community. Community structure based on cyclic activities of daily or seasonal basis is called temporal structure of community. Ecosystems where layering is present are have spatial structure. Different populations are present at different layers of forests. Variety of animals present in a community is called **community diversity**. It is also called community richness. Stable climate, more resources, high productivity, moderate predation promote diversity.

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LESSON226

Community Stability

Communities are born and they die. Some changes in community are not permanent. They are replaced by next generation. Whereas some changes are permanent.

It is the sequence of community types during maturation of an ecosystem is called ecological succession. It is the change in community structure in a predictable way. It is the first community established in an area devoid of life is called pioneer community. e.g., a lake formed by glacier and the first community establish in this lake will be **pioneer community**. During ecological succession and ecosystem is occupied by different communities in a sequence and each successional stage is called a seral stage. The final community during an ecological succession is called **climax community**. They are complex and have more species diversity.

The process of succession in an area that was not supporting any life previously is called **primary succession**. E.g., Ecological succession in a lake formed by glacier or land exposed by land sliding. The process of succession where previous life destroyed by some natural catastrophs (e.g., fire) is called **secondary succession**.

LESSON227

ECOLOGICAL NICHE

It is the role of an organism in an ecosystem which include all the attributes of life style of an animal i.e., where it lives? what it eats? where it search for food? what physical parameters it prefers?

Ecological niches of two species may overlap. This will result a competition between them. Animals will compete for same resources. Competition between species may be reduced if resources are divided. One part of resources is used by one species and second part by other species. This mechanism is called **niche diversification**.

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LESSON228

FOOD CHAINS

It is the sequence of organisms in an ecosystem where energy moves from one organism to the next organism. It tells us the sequence to eat and being eaten. Food chain is important: i). To understand the feeding relationships between organisms. ii). To explore the energy flow mechanisms and matte circulation. Iii). To understand the problems of biomagnification.

Food chains are mainly of two types: grazing food chain and detritus food chain.

a. Grazing food chain

It is the food chain that starts from the living green plants. Green plants are eaten by herbivores which are eaten by carnivores. Such food chains are directly dependent on

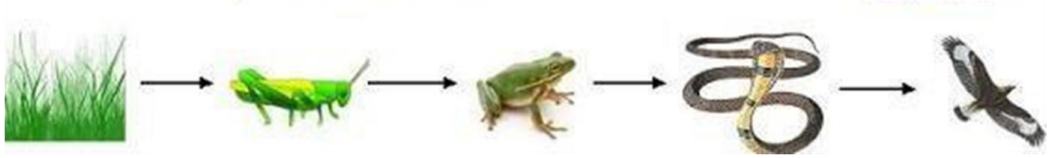


Fig. Grazing food chain of an ecosystem

sunlight.

Detritus Food Chain

It is the food chain that starts from dead organic matter. Organism of detritus food chain are

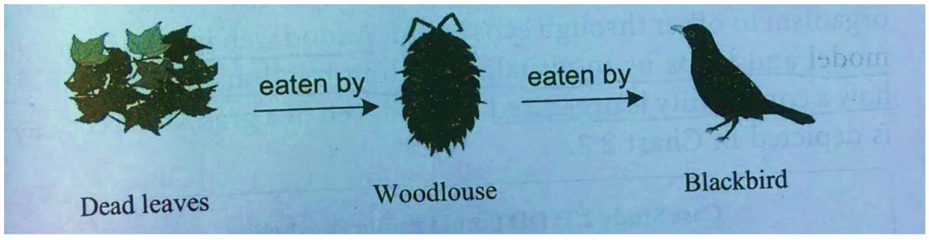


Fig. Detritus food chain of an ecosystem

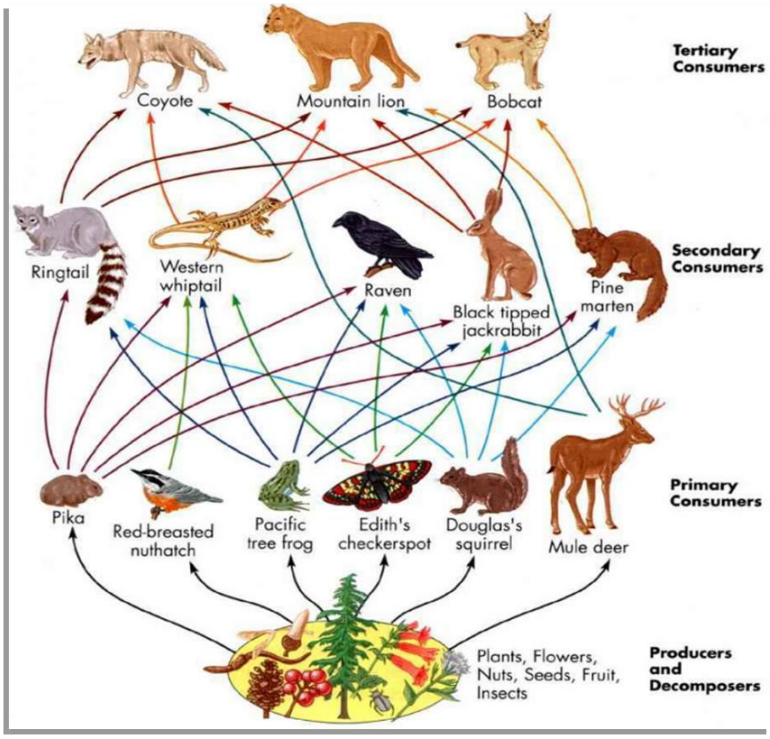
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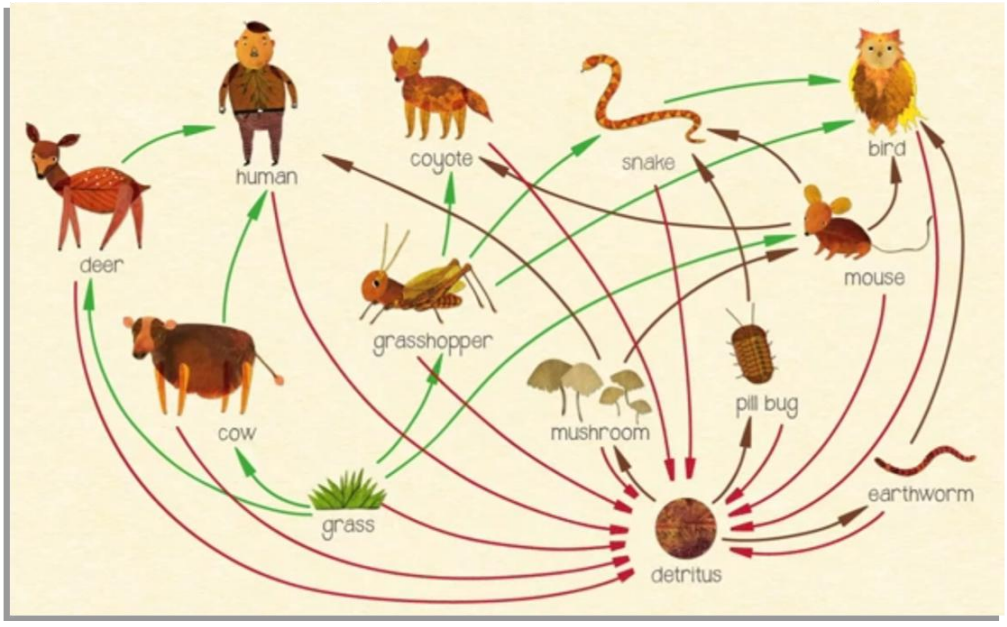
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LESSON229

Food Webs

It is a complex network of organisms through which energy and matter is transferred in an ecosystem. Food web consists of many interconnected food chains. Grazing food webs are the food webs where plants make the starting point. These food webs based on algae and green plants. The detrital food webs are the food webs where decomposers make the starting point. Bacteria and fungi move the food material to the carnivores.





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LESSON230

TROPHIC STRUCTURE

As food webs are very complex. The animals feeding in a food web are distributed into different trophic levels. Autotrophs make 1st trophic level.

a. Producers

These are the autotrophs which obtain nutrition from inorganic compounds. They form the first trophic level of an ecosystem. Producers are mainly of two types: Photoautotrophs and chemolithotrophs. **Photoautotrophs** use sunlight as energy source for the synthesis of food from inorganic compounds. They include green plants, algae and photosynthetic bacteria. Photosynthesis transform sunlight into chemical energy. Out of total sunlight reaching to earth 30% reflects back, 20% is absorbed by atmosphere, 50% absorbed by earth and less than 1% of sunlight is used by photosynthesis. **Chemolithotrophs**obtain energy by oxidizing inorganic compounds. They usually live in extreme environments. e.g., sulfur bacteria.

LESSON231

a. Consumers

Organisms which cannot synthesize their food from inorganic compounds. They eat other organisms to get nutrition. All animals are consumers. Those organisms which feed on plants are called **primary consumers**. They are also called herbivores. E.g., insects, grain eating birds. Different reptiles, birds and mammals which feed on leaves and other vegetation are also herbivores. Those animals which feed on herbivores are called **secondary consumers.** They are also called carnivores. e.g., lion, tiger and frog. Those animals which feed on secondary consumers are called **tertiary consumers.** e.g., a snake when eating a frog, a jaguar when hunting for a crocodile.

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MODULE 232

a. Decomposers

These are the organisms which obtain energy by breaking down dead organic matter into simpler compounds. They are also called saprotrophs. Chief decomposers of an ecosystem are bacteria and fungi. They release extracellular enzymes which breakdown substrate molecules. Organisms at all trophic level when die are used by decomposers. If there are no decomposers mounts of dead bodies will be everywhere. All useful minerals will remain lock in dead bodies. There will be little space to live. Decomposition also recycle the nutrients in ecosystem.

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LESSON233

ECOLOGICAL EFFICIENCIES

It is the efficiency by which animal at a trophic level convert food into new biomass. A large amount of food is not digested by the animals. Herbivores have lot of plant biomass, but they are less efficient. Carnivores are more efficient. At an average there is 10% biomass conversion efficiency at each trophic level. Biomass conversion also depends upon the maintenance requirement of animals. Efficiencies range for less than 1% for most herbivores to more than 35% for ectotherm carnivores.

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LESSON234-235

ECOLOGICAL PYRAMIDS

It is the graphical representation of relationship between organisms at different trophic levels of an ecosystem. There are three most common types of pyramids: Pyramid of number, pyramid of biomass and pyramid of energy.

a. Pyramid of Numbers

It is the graphical representation of total number of different organisms present at each trophic level. This also shows the population of organisms in a food chain.

For example in a grass land the survey was conducted to check the trophic structure. The number of producers was 5.842 x 10^6, primary consumers 7.08x10^5, secondary consumers: 5.4x10^4 and tertiary consumers was one.

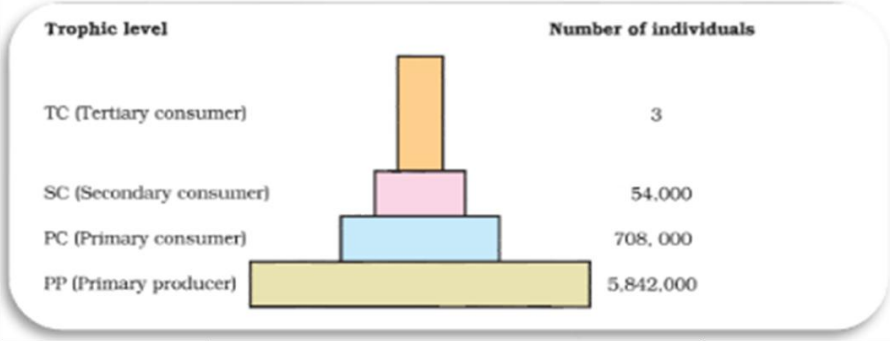


Fig. Pyramid of number

b. Pyramid of Biomass

It is the representation of total mass of living matter present at each trophic level. Biomass decrease at each ascending trophic level. Biomass is measured in grams of dry mass per square meter. For example in a pond the survey was conducted to check the trophic structure. The biomass of producers was 809 g/m^2, primary consumers was37 g/m^2, secondary consumers was 11 g/m^2 and tertiary consumers was 1.5 g/m^2.

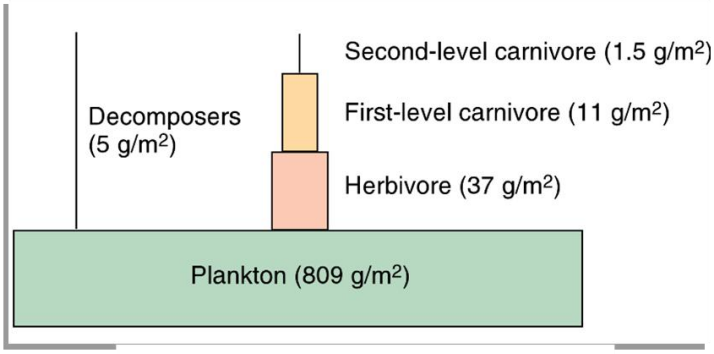


Fig. Pyramid of biomass

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LESSON236

ENERGY AND MATTER FLOW IN AN ECOSYSTEM

1. Energy Budget

It is the balance sheet of energy gained and energy expenditure of an organism. An organism is an open thermodynamic system which exchange energy with its surrounding in three ways *i.e.*, heat, work and potential energy of biomolecules. Energy budget allocation varies for ectotherms and endotherms. Ectotherms rely on environment as heat source while endotherms rely on metabolic heat to maintain body temperature.

When an animal eats, food passes through its gut and nutrients are absorbed. Total amount of energy assimilated by an organism when an animal eats food is called **gross productivity**. Most of the energy assimilated is used in metabolic demands which include cellular metabolism and regulation of body temperature. The amount of energy stored in the form of biomass is called **net productivity**.

4. Energy Flow

Organisms require energy in one form or the other. Energy flow in an ecosystem is major factor for survival for great number of organisms. Energy flows via food chain and food web. Source of energy is sunlight. Normally 10% of energy flows to next trophic level. Energy flow is unidirectional.

All elements essential for life are derived from environment. They make all form of molecules for life. When animals die and their dead bodies are decomposed, these materials are recycled in the ecosystem. Nutrient (material) flow in ecosystem in a cyclic way.

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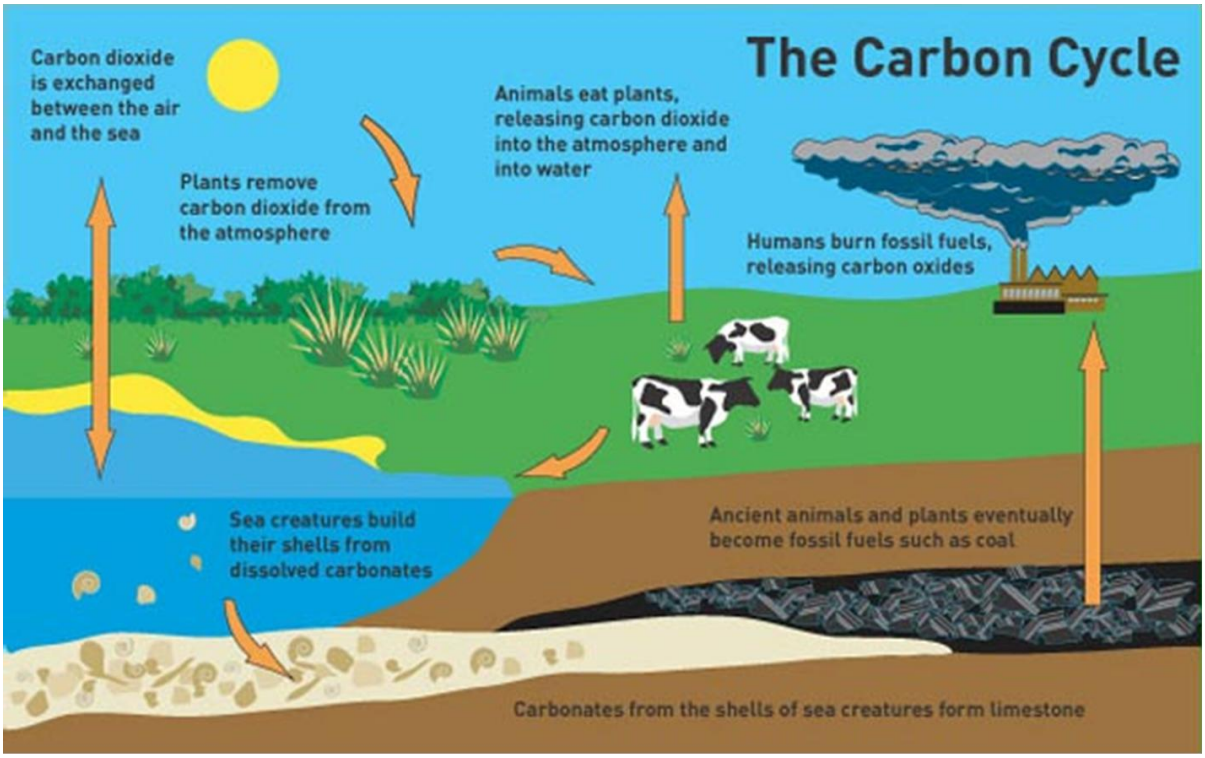
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LESSON237-238

CARBON CYCLE

Carbon is the basic element of all organic molecules. All biomolecules have a carbon backbone. Carbon has ability to bond with four other atoms at the same time. Carbon is plentiful on earth and is rarely limiting factor. Reservoir for carbon is CO_2 in atmosphere and water. Photosynthetic plants use this carbon dioxide of air and convert organic compounds. These plants are being eaten by animals. Animals utilize these organic compounds for the synthesis of different biomolecules and also use as a source of energy. When organic matter is completely oxidized, it releases CO_2 into atmosphere. In aquatic environment, CO_2 combines with water to form H_2CO_3. When CO_2 level in water decreases H_2CO_3 breaks to release CO_2. Some carbon is tied up in the form of calcium carbonate ($CaCO_3$). This $CaCO_3$ is in the part exoskeleton and endoskeleton of mollucs, echinoderms, and other invertebrates. They ultimately form limestone reservoirs.

A large supply of carbon is also tied up in fossil fuels. During 65 million year of Carboniferous period (280mya), organic matter got buried and converted into coal, oil and gas. When fossil fuels are used, carbon returns to atmosphere.



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LESSON239-240

NITROGEN CYCLE

Nitrogen is an essential component of proteins, nucleic acids and ATP. It is also found in membrane lipids and modified carbohydrates. Atmosphere is main reservoir nitrogen. About 78% nitrogen present in air, but this is not directly available to plants. Plants can use nitrogen in the form of NH_3 and NO_3^-. Animals obtain nitrogen from plants. Atmosphere nitrogen converted to NH_3. This process is called **nitrogen fixation**. Aqueous form of NH_3 is NH_4^+. Nitrogen fixation is mainly done by bacteria in the roots of leguminous plants and cyanobacteria. Lightening is another source of nitrogen fixation. In industry such as fertilizer industry nitrogen fixation is done by artificial method. In the process is **nitrification** during which NH_3 is converted to nitrate (NO_3^-). It is a two step process. In first step NH_3 is converted to nitrite (NO_2^-) and in second step NO_2^- is converted to NO_3^-.*Nitrococcus* and *Nitrobacter* do this job, respectively. Plants use these nitrates to synthesize proteins and nucleic acids acid. Some plants have ability to use ammonia (ammonium ions) as nitrogen source. Animals utilize plant materials as a source of nitrogen. The process by which nitrogen is incorporated into plants and animals is called assimilation. When plants and animals die or release nitrogenous waste, this organic nitrogen is also converted into ammonia by a process called **ammonification**. Ammonia released in this process may be directly assimilated by the plants or used after nitrification. Some bacteria convert nitrates back to nitrogen is called **denitrification**. This process is completed under anaerobic conditions. *Pseudomonas* and *Aachromobacter* are the examples of denitrifying bacteria.

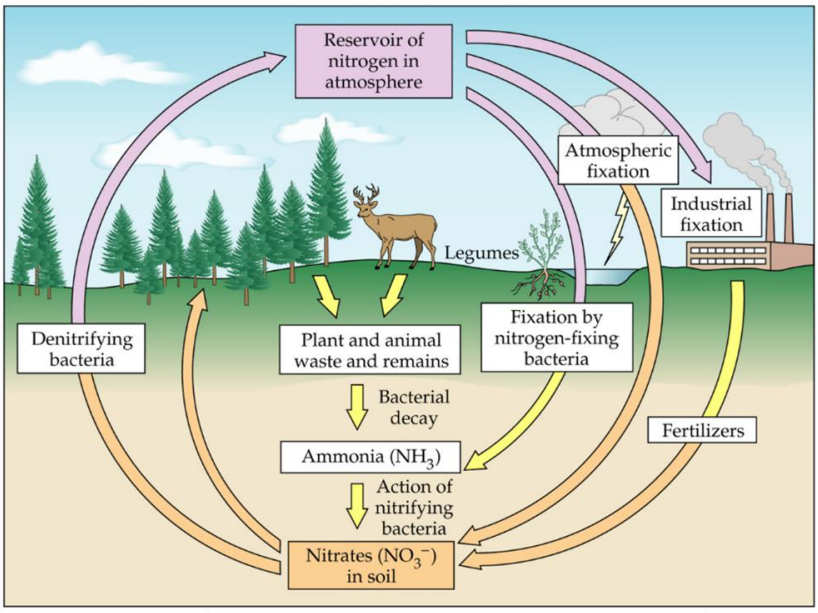


Fig. The nitrogen cycle

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LESSON 241

PHOSPHORUS CYCLE

1. Importance of Phosphorus

Phosphorus is an essential component of DNA, RNA, ATP, Phospholipids, Vertebrate skeleton and Teeth. Sedimentary rocks contains deposits of phosphorus. Extraction of phosphorus takes place by weathering, rain and other erosion processes. Plant, microbes and fungi absorb phosphorus from soil. Animals use phosphorus by utilizing plants. Some phosphorus also comes through water. As plants and animals die, phosphorus is return back to the environment via decomposition. Living organisms may again use this phosphorus. Some phosphorus may run along with water and become part of sediments. Human has significant impact by using phosphorus in fertilizers. Agricultural runoff takes phosphorus to water bodies which excessive growth of algae. This phenomenon is called eutrophication.

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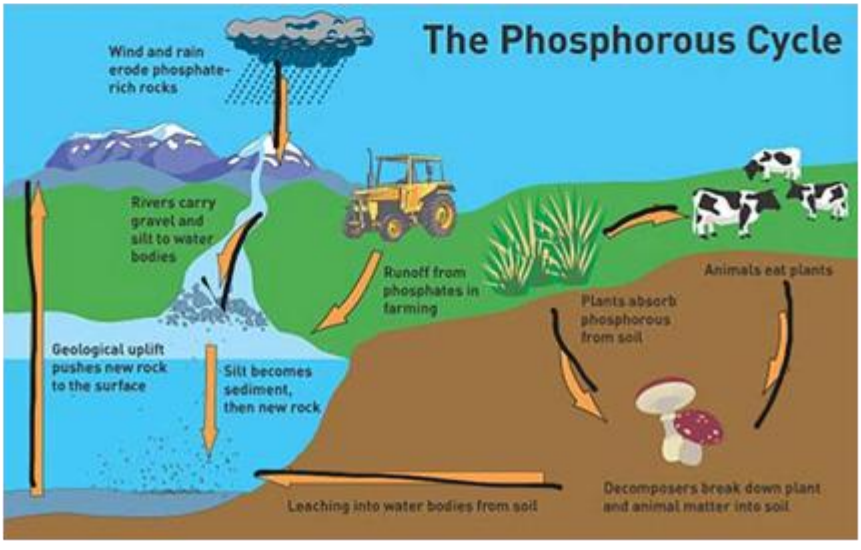
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LESSON 242

SULFUR CYCLE

Sulfur is an essential component of proteins, enzymes and vitamins. It is abundant in nature and is rarely a limiting factor. Sulfur is a bright yellow crystalline solid. Most of it is tied up in rock and buried in depth of oceans. Atmosphere also contain oxides of sulfur. The process by which organic sulfur is converted into inorganic form i.e., hydrogen sulfide, elemental sulfur and sulfide minerals is called mineralization and the step at which sulfides and elemental sulfur are converted to sulfates is called oxidation. Sulfates are reduced by plants, fungi and bacteria and incorporated

Fig. Phosphorus cycle



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LESSON 243

WATER CYCLE

Water is essential for life. Life cannot exist without water. About 97% water is marine with high salt concentration, 3% freshwater, mostly glaciers and groundwater. Less than 1% of water is available to human. Water in the oceans, lakes, swamps, rivers and soil when heated is turned into vapors. Similarly water also evaporates from surfaces of plants and animals. The loss of water from plants is by transpiration. Water also evaporates directly from ice by the process of sublimation. Water vapors move up higher and condense to form clouds. Clouds are pushed

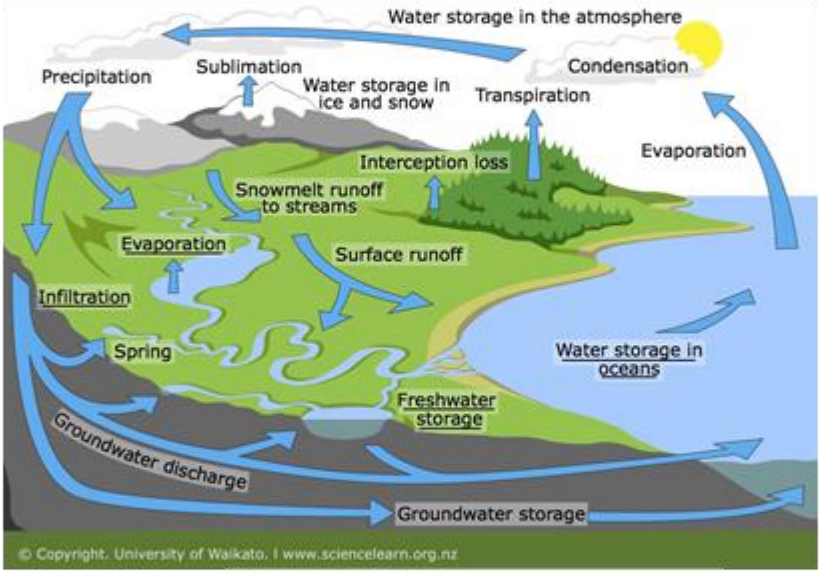


Fig. The water cycle

around the world by moving air. Further cooling causes precipitation. Precipitation may be in the form of rain, hail or snow. About 78% of precipitation occurs over oceans. Rain water flows across the land is called runoff. It includes: Surface runoff and channel runoff. Flow of water from surface into ground is called infiltration. It becomes soil moisture or ground water. Underground water is present in aquifers. This water may return to surface as springs or by pumps. Under ground water moves slowly and may remain in aquifer for thousands of years. Eventually is reaches to the oceans and become part of ocean water.

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LESSON 244

BIOSPHERE AND ITS SUBDIVISION

Biosphere is thin outer layer of earth which can support life. It includes all life on earth and the physical environment on which living organisms exist and interact. Life exists in soil, water and air. Biosphere is subdivided accordingly: Lithosphere, hydrosphere and atmosphere.

Lithosphere is the rigid material on the surface of earth. Most of the terrestrial animals live on lithosphere. Lithosphere is the ultimate source of all mineral elements required by all living things. **Hydrosphere** is the water on the surface of earth. It extends into the lithosphere and atmosphere. Water is distributed on earth by global hydrological system. Water returns to oceans by precipitation. **Atmosphere** is the gaseous content of biosphere. Atmosphere extends up to 3500 km above the surface of earth. Life in atmosphere is present in lower 8-15km called troposphere. The main gases in troposphere are nitrogen, oxygen and carbon dioxide.



Fig. Biosphere: atmosphere, hydrosphere and lithosphere

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LESSON 245

ECOSYSTEM AND BIOME

Ecosystem is the interaction between living and nonliving components of the environment while an area of earth with distinctive association of plants and animals is called biome. Major biomes of the world include: tundra, coniferous forests, deciduous forests, grasslands and deserts. Botanists divided terrestrial environment long ago into different biomes. Characterization of biome on the basis of animal life is difficult. A biome is identified by dominant plant formation. Because plants support a particular type of animals, so the biomes also have characteristic type of animals. The boundaries of biome are not distinct and contain trees of both types of biomes. This indistinctive boundary having a mixture of plants is called **ecocline**.

Biomes are distinct by climate, pattern of rainfall, temperature and solar radiations. Warm air at equator rises and move towards poles and cold air away from pole. The earth rotation deflect this air to right and left of the poles. The effect is called **Corioliseffect**. Air circulation in each hemisphere is broken into three zones from poles: Zero to 30° latitude, 30° to 60° and when cold air move back to poles at high altitude.

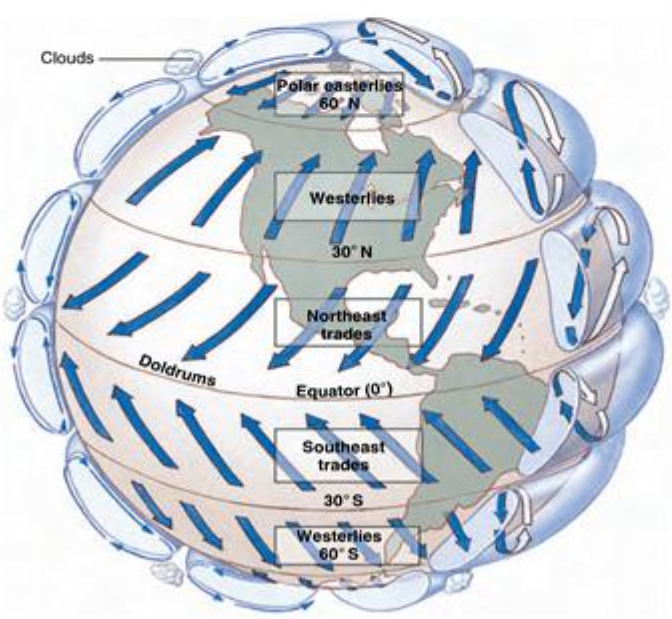


Fig. Corioliseffect and cells

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LESSON 246-247

TUNDRA ECOSYSTEM

Treeless region found in arctic and top of mountains with severe cold climate and permanently frosted subsoil. Tundra havea long winter interrupted by a short summer. Plant can grow in this summer of 60 days. Soil remains frozen in most of the year. Subsoil is called permafrost, as it is permanently frosted. Most tundra region is covered with bogs, marshes, ponds and decayed vegetation. Average precipitation is very low approximately less than 25cm/year. Average annual temperature is -10°C although it may drop to -50°C in winter and rise to 10°C in summer. There are no trees, plant life is mainly represented by lichens, grasses and dwarf woody plants. Growth of plants is very slow due to extreme conditions. Animal life is mainly represented by rodents such as arctic hare, musk oxen, arctic foxes, owls, weasels and caribou (reindeer). Tundra exists in two forms: Arctic tundra and Alpine tundra

Arctic tundra is present in northern hemisphere. It includes the areas of northern Russia and Canada. It is the home of Nganasan and Nenets people. Temperature ranges form -30°C to 20°C. Animal life include arctic fox, polar bear, gray wolves, caribou, snow geese and musk ox. Plant life lichen, mosses, grasses and shrubs. **Alpine tundra** is present athigh elevation are too cold and windy to sport plant life. North American Cordillera, the Alps and Pyrenees of Europe, the Himalaya and Karakoram of Asia, the Andes of South America, and the Eastern Rift mountains of Africa. Flora of alpine tundra includes Lichens, grasses and shrubs. They have much more roots and rhizome than shoots which are also storage site during winter. Animal life include kea, marmot, mountain goat, bighorn sheep, chinchilla, yak and pika. Arctic tundra is changing dramatically due to global warming. Animal previously not found in arctic tundra are now competing with native animals. Shrubs are getting taller and animal like spider are more active. Permafrost is melting. It holds large amount of carbon and methane. Climate change may change tundra from carbon sink to carbon contributor.

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LESSON 248-249

CONIFEROUS FORESTS

Coniferous forest are dominated by cone bearing, needle shape evergreen plants called conifers. They have cool summer and cold winter. Growing season is 130 days. Soil is acidic and relatively infertile. There is moderate rainfall ranging from 40 to 100cm/year. Most precipitation in the form of heavy snow. Average annual temperature is 10°C although it may drop to -40°C in winter and rise to 20°C in summer. Plant life is represented by spruce, fir and pine. They are evergreen. Some primary production also on warmer summer days. Animal diversity is low but large population. Major animals are snowshoe hare, lynx, wolves, caribou, moose and summer moths.

Coniferous forests are evergreen, their primary productivity is higher than grasslands but lower than deciduous and tropical forests. Decomposition process is slow due to cool environmental conditions. Nutrients are locked up in pine needles that fall from trees. Fire is important in recycling of nutrients and promoting seed germination. Coniferous Forests are thelargest terrestrial biome on earth. Stretched from North America to Europe and Asia in northern hemisphere is called taiga. Whereas, temperate Coniferous Forests are present worldwide at high altitudes in America, Africa, Europe South Asia. In Pakistan they are present at 1000 to 4000 altitude. Coniferous forests are present in Chitral, Swat, Dir, Abbot Abad, Hilly Areas of Balochistan, Murree and Azad Kashmir. Coniferous trees are used in buildings and furniture. So, these forest are seriously damaged by human. Climatic changes are affecting the coniferous forests by mild, short winter and acid rain.

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LESSON 250-251

TEMPERATE DECIDUOUS FORESTS

These are forests with moderate climate and well define summer and winter. They drop their leaves during autumn. These forests have four seasons: Winter, spring, summer and fall. Annual precipitation ranges between 75 to 150cm/year and temperature varies from -30°C to 30°C. Soil is fertile and brown in colour. Decomposition of fallen leaves enrich the soil due to which primary productivity is higher than grasslands and coniferous forests. Plants are broad leaves trees. There is great diversity of animal life. Important animals include spiders, ants, white tailed deer, moose, wolves, raccoons bears and tigers. Temperate deciduous forest are located in Eastern USA, Canada, Japan, China and some part of Australia. Human get valuable timber from these forests. These forests have been mostly cut down from Europe for farming purposes. Some have been converted to commercial forestry.

LESSON 252-253

THE GRASSLANDS

These are the areas where vegetation is mainly dominated by the grasses. They account for 20-40% of world land area. They lie between forest and deserts. Grassland is found where there is not enough rain fall. Precipitation is 25 to 100cm annually and temperature ranges from -20°C to 30°C. The upper layer is nutrient rich making them ideal for agricultural use. But the low water availability may eventually convert them into deserts. Grasslands of North America are called prairies while the grasslands of South America and Africa are called tropical savanna. **Praries**are treeless while **savanna** has scattered trees (less than 30%).

Plant life is dominated by different types of grasses making layers. Plant life include clover, salvia, wheat, barley, coneflowers and many species of grasses such as, purple needle grass, wild oats, foxtail, ryegrass, and buffalo grass. Grasslands support large populations of herbivores and carnivores including insects, deer, zebra, wildebeest, giraffe, bison, lions, tigers and hyenas. Important grasslands include Great Plains of North America, Pampas of South America, Veldt of South Africa, Steppes of Central Eurasia, and surrounding the desert of Australia. Continuous global warming has great impact on grasslands. Grasslands are under extensive pressure due to conversion to agricultural land and urban development.

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LESSON 254-255

THE CHAPARRAL

It is the biome fount in small patches in most of the continents. It is characterized by very hot and dry summer with temperature up to 37.5°C. Winter is very mild and usually about 10°C and most of the rain occur in winter. Annual precipitation is 30-75cm. Soil is thin, rocky, nutrient poor, highly susceptible to erosion and contains low amount of moisture. Plants of chaparral are low and shrubby. They have tough and waxy leaves which are broad leaves and evergreen. Plants have no thick bark so not very resistant however their recovery is very fast. There are two types of plants: sprouters and seeders. **Sprouters** aredormant meristem, storage and intact root help to regenerate quickly while **seeders** arelong lived seeds, germination induced by fire. Plants include toyon, chamise, poison oak, scrub oak, yucca and sumac. Important animals are insects, rodents, rabbits, lizards, snakes and mule deer. Chaparral is represented by west coast of United States and South America, South Africa, western tip of Australia and coastal area of Mediterranianean.

LESSON 256-257

DESERT ECOSYSTEM

It is a region which receives annual precipitation less than 25cm. They are characterized by scarce vegetation and extreme temperature occilation. Extremely dry climatic conditions are present here. It has lowest precipitation in comparison with other biomes. There go years and years without precipitation. At day time temperature may be 38 while at night it may be 4°C. Desert soil is dry with low organic nutrients, low fertility and low water holding capacity. Only those plants can grow which ability to survive in extreme conditions of drought. Reduced leaves, succulent stems, deep stomata and waxy cuticle are the important adaptations of desert plants. Predominant plants in desert ecosystem are different species of cacti and thorny shrubs. They have reduced foliage and drought resistant seeds. Animals of desert include different types of arthropods, reptiles (snakes, lizards), birds (road runner, cactus wren, turkey vulture and burrowing owl) and mammals (mule deer, jackrabbit, kangaroo rat and ground squirrel).

Desert is only ecosystem which is increasing with human activities. This is because the grasslands used for agriculture and ultimately converted into desert land. Important deserts of world are Antarctic Desert, Arctic Desert, Sahara Desert, Australian Desert, Arabian Desert, Gobi desert, Kalahari Desert, Thar Desert, Columbia basin and Dasht-e-kavir.

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LESSON 258-259

TROPICAL RAINFORESTS

It is a hot moist biome found near the equator. They are the richest in biodiversity and most exciting areas of the earth. Tropical rainforests have warm temperature and abundant rainfall. Average temperature ranges from 20°C to 25°C and it remains constant. Annual rainfall is more than 200cm. There is little seasonal variation in day length. Soil is relatively nutrient poor. Most of the nutrients are tied up in organisms. High moisture and temperature make nutrient recycling rapid.

Tropical rainforest have 50% of plants and animals diversity with just 6% land. Plant life is very diverse. There are thousands of plants species. However, none of them dominating. Trees are broad leaved and non-deciduous. Plants in tropical rainforest are highly stratified. A single hectare typically contains 50 to 70 species. Some of the important plants are palm species, rubber tree, bamboo, orchid, tualang, different epiphytes and pitcher plants. Both soft bodied and hard bodied endothermic animals are abundant. Important animal groups are annelids, arthropods, amphibians, reptiles, birds and mammals. Tropical rainforest are located in South and Central America, Western Africa, Southeast Asia and Australia. Once they were 14% and now they are 6%. It is estimated that in next 40 years the remaining will be lost as well. The deforestation mean habitat loss, extinction, soil erosion and climate change.

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LESSON 260

FRESHWATER ECOSYSTEMS

Freshwater ecosystems are classified according to water flow.

Lotic Ecosystem

It is the aquatic ecosystem of flowing water. It includes brooks, streams and rivers. Attached aquatic plants and algae provide some of the primary productivity. Most of the primary production comes from the surrounding plants by falling their materials during autumn and rainfall. Lotic systems typically have high oxygen level. Turbulence of water while flowing over rocks, logs and other objects causes oxygenation. This support the animal life in lotic system. Mainly algae and attached aquatic plants are indigenous producers. Animals include different aquatic invertebrates, larvae of insects, fishes, amphibians reptiles birds and mammals. Water Currents are the most important physical factor of lotic habitat. Animal adaptation include dorsoventral surface, strong swimmers, burrowing and Attachment with substrate.

LESSON 261

Lentic Ecosystem

This is the ecosystem of standing water. Lentic ecosystem include lakes and ponds. Vertical stratification is common especially in temperate region. In summer, upper layer is warm while lower layer is cold and more dense, so no mixing. In fall, upper layer is dense so mixing of water occurs. Photosynthesis mainly in upper layer of water as light is filtered by water. Phytoplankton are main producers of lentic ecosystem. Deeper layers are depleted of oxygen in summer. In fall the mixing of entire water column supply oxygen to deeper layers. Deep, cold, nutrient poor lakes are relatively unproductive called as oligotrophic lakes. Shallow warm, nutrient rich lakes are highly productive called as eutrophic lakes. Plant life include phytoplankton, periphyton algae, emergents, floating- leaves, submersed and free floatingplants. Animal life include zooplankton, invertebrates like crabs, crayfish and shrimps, molluscs while vertebrates mainly includes variety of fishes, reptiles and birds.

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LESSON 262

MARINE ECOSYSTEMS

Most important character used to distinguish the marine water from freshwater is salinity. Marine water typically contains 3.5% salts. Marine ecosystems are of following types.

Estuaries

Estuaries occur where freshwater meets the seawater. This dilutes the salinity of seawater. Estuaries are highly productive because the streams carry nutrients of freshwater and tides bring nutrients from ocean. Producers of estuaries include phytoplankton, attached aquatic vegetation and emergent vegetation such as cordgrass. Estuaries are nurseries for crustaceans, molluscs and marine fishes. They are the feeding grounds for wading and swimming birds. Human have drained and filled estuaries. Estuaries are polluted by oil exploration and drilling. Acids rain have also threaten the productivity of estuaries.

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LESSON 263

The Intertidal Zone extends from splash zone ocean to low tide marks. This region is subjected to the action of both tides and waves. It is the harshest and the richest of all marine environments. Here is pounding surf, sun, wind, rain, erosion, sedimentation and temperature fluctuation. This zone has bounty of habitat and plenty of nutrients. Here, plants are larger. They include seagrass, red algae and mangroves. Motile animals like fishes and crabs move with tides. Other such as amphipods, isopods, ghost crabs and polychaetes burrow in the sand and mud. Water trapped in tidal pools support a rich animal life such as echinoderms, barnacles, limpets and chitons. Attached animals close shells to avoid drought during low tides.

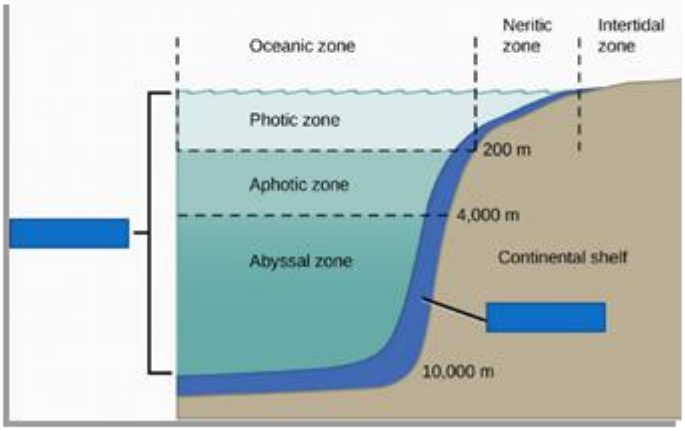


Fig. Different zones in marine ecosystem

LESSON 264

The Neritic Ecosystem extends from littoral zone to the edge of continental shelf. It is also called the shallow water. Nutrients are delivered by rivers and upwelling at the edge of continental shelf. This region is more productive than open sea. Planktonic and attached algae contribute to the primary productivity. Plants include kelps, algae, seaweeds, seagrasses and phytoplankton. Invertebrate animals include crabs, sponges, echinoderms, corals, and polychaetes whereas, vertebrate fauna include different types of fishes such as code, tuna, herring, capelin and many other species.

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LESSON265

Oceanic Ecosystem starts from continental shelf. It is the open sea. As it is nutrient poor so, it is relatively unproductive. Virtually all primary productivity is in the upper water. Phytoplankton in the upper water are the main producers of oceanic system. Invertebrate animals include worms, echinoderms and molluscs. Vertebrates include fishes from small fishes to whales and sharks. At depth no light so the animals here have luminescent organs which serve as torch light. These animals depend on nutrient coming from upper layers.

LESSON 266

Coral Reef is the underwater ecosystem. They are formed by colonies of stony corals. Coral are members of class Anthozoa. They grow in association of zooxanthellae. Polyps of corals are held together by calcium carbonate. Reefs are formed over thousands of years in warm, clear and relatively shallow waters. The greatest depth for their growth is 50-60 m. Below this light is insufficient to support zooxanthellae. Coral Reefs are the richest source of biodiversity in oceans. Coral reefs occupy less than 0.1% of oceans, yet they are home of more than 25% marine species. Residents include sponges, cnidarians, nematodes, crustaceans, molluscs, echinoderms, tunicates and fishes.

LESSON 267

ECOLOGICAL PROBLEMS

Human Population Growth

Population growth is the increase in number of individuals in a population. Expanding population is the root cause of most environmental problems. More population means more need for resources i.e., food, urbanization and industrialization. The present population of earth is close to more than 7 billion. Virtually all scientists agree that even 5 billion is more for earth to support with all facilities. Maximum carrying capacity of earth is may be 9 to 10 billion people.

Age structure shows the proportion of individuals at prereproductive, reproductive and postreproductive stage. Developing countries have more infants means high birthrates. In contrast to human, in natural population high birthrate is accompanied with high mortality.

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LESSON 268

Pollution

Pollution is any detrimental change in ecosystem. Mostly is results from human activities. More human population<more demands<pollution. Toxic waste of industries, sewage waste and agricultural runoff are the important sources of water pollution. Burning of fossil fuels release CO_2, oxides of sulfur and nitrogen and Increased CO_2 are polluting air. This this increasing the threats of acid deposition, smog and global warming. Land pollution is mainly caused by oil spillage, plastics, pesticides, toxic industrial waste, garbage and buildings leftover.

Wastes and poisons enter in food chain and travel to the organisms at highest level of food chain including human. They are not biodegradable and interrupt physiological processes. They accumulate in the bodies of animals and cause neurological, developmental and different types of physiological disorders.

LESSON 269

Resource Depletion

Materials or substances occurring in nature which can be exploited for human benefits are called natural resources. Important natural resources are fossil fuels, minerals, air, water, soil and biodiversity. As human population is increasing rapidly and resources are limited. For better life man is overexploitation natural resources. This will ultimately use of pollute the natural resources and human survival will be at a great risk.

LESSON 270

Biodiversity Loss

Current documented biodiversity of earth is more than 1.6 million. Estimate is 4 to 30 million. More accurate is between 8.7±1.3million. Biodiversity of all natural areas is threatened mainly due to habitat loss, acid rain, pollution, urbanization and agriculture. Human is exploiting 40% of earth net primary production.

Important solutions are realization about biodiversity. 22nd May is World Biodiversity Day. There is the need for a system of international ethics that value natural diversity.

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Remember me in your prayers.

Regards: Aiman zoologist